



Small signal stability analysis for different types of PMSGs connected to the grid



Da Xie ^{a,*}, Yupu Lu ^a, Junbo Sun ^a, Chenghong Gu ^b

^a Department of Electrical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China

^b Department of Electronic and Electrical Engineering, University of Bath, Bath BA2 7AY, UK

ARTICLE INFO

Article history:

Received 3 January 2016

Received in revised form

5 December 2016

Accepted 9 January 2017

Available online 11 January 2017

Keywords:

Converter system

Eigenvalue analysis

PMSG-based wind turbine

Small-signal stability

Two-mass shaft model

ABSTRACT

Small signal stability of permanent magnet synchronous generator (PMSG)-based wind turbines connected to the power grid should be studied properly in order to facilitate damping strategy design. In this paper, unified small-signal models for different types of PMSGs are developed to study their small-signal stability. The models are composed of mechanical systems, electrical systems and control systems. A two-mass shaft model for the mechanical system is provided to analyze the dynamic and steady-state behaviors of the wind turbine and generator rotor. Meanwhile, PMSG, converter system and transmission line are separately modeled to build unified small-signal models for three PMSG-based wind turbine generator systems (WTGS). Then, based on unified small-signal models, eigenvalue analysis is conducted to determine the relation between different oscillation modes and state variables through calculating participation factors. With modal analysis, the developed small signal models are able to find out all types of oscillation modes for PMSGs connected to the power grid, which are subsynchronous oscillation (SSO), subsynchronous control interaction (SSCI) and low-frequency oscillation, including frequency and damping of each oscillation mode. Different initial values of the small signal models can influence both frequencies and damping ratios of oscillation modes, which lay basis for further damping strategy study.

© 2017 Elsevier Ltd. All rights reserved.

1. Introduction

As a prospective renewable energy source, wind power experienced substantial growth from 2010 to 2011 with a total capacity of 238GW worldwide. In addition, more than 20% of global electricity is predicted to be supplied by wind power by 2030 [1]. Various topologies of wind turbines have been developed, including fixed speed induction generator (FSIG), doubly fed induction generator (DFIG) and permanent magnet synchronous generator (PMSG) systems [2–4]. The large-scale wind turbine generators (WTGs) connected to the power grid are raising severe concerns over oscillations, which decreases the stability of wind farms.

PMSG-based WTGs are widely adopted in offshore wind parks because of their high efficiency, low maintenance costs and high reliability [5]. For grid-connected WTGs, torsional vibrations of mechanical systems can be excited by grid faults. The vibrations may lead to shaft faults if the mechanical damping cannot compensate the negative damping of electrical systems [6]. In Ref. [7], a lumped mass model of an FSIG is developed and simulation results show that the FSIG system is excited by wind speed

and gear mesh stiffness fluctuations. In Ref. [8], simulation results indicate that torsional vibrations can be potentially mitigated for DFIG with the de-coupled control of generator speed and terminal voltage. In Ref. [9], a coupled electromechanical model of a PMSG-based wind turbine is presented and it is found that proper rotor construction can effectively reduce mechanical vibrations. Above literature is concentrated on torsional vibrations of the mechanical systems of WTGs, but the influence of electrical and control systems is neglected.

Conventional classification is based on the frequency band. Traditional theories assume that power oscillation is caused by the negative damping effect of generator fast excitation system, with oscillation frequency generally between 0.2 and 2.5 Hz. By contrast, the interactions between wind turbines and the power grid only gain research attention in recent years, where these interactions are divided into SSCI, SSR, SSO and low-frequency oscillations [10,11]. According to IEEE Committee Report [12–14], SSCI is called sub-synchronization control interaction, which is closely related to high-speed control of power electronic system and series compensation capacitor, i.e. the electrical resonance frequency. Normally, it appears under situations with lower series compensated degrees [15,16]. SSR is caused by a series of capacitors in the grid, while SSO is a type of stability problems of high-speed

* Corresponding author.

E-mail address: profxyzg@hotmail.com (D. Xie).

electronic switch control systems, mainly caused by the positive feedback from the improper unexpected phase relationship. SSR and SSO affect the shaft mechanical system, with their frequencies close to the free frequencies of the shaft system [17–20]. Low-frequency oscillation in PMSG generation system is considered to be caused by the negative damping effect of flux linkage of rotor windings [21,22]. However the low-frequency oscillation mechanism for PMSG is still unsettled in theory.

Eigenvalue analysis on small-signal models of WTGs is an effective way to analyze all the mentioned oscillation modes. The relative participation of system components in oscillation modes is measured by participation factors [23]. The sensitivity analysis on eigenvalues can also facilitate parameter design for WTGS. Because of the distinct advantages, a significant bunch of research has been focused on small-signal stability analysis of FSGs and DFIGs. In Ref. [24], based on the small-signal model of an FSG, a frequency response approach is used to evaluate the sensitivity of system modes in response to system parameters. In Ref. [25], a complete dynamic equation of an FSG system is derived and the relation between its dynamic modes and state variables are studied through eigenvalues analysis. In Ref. [26], a DFIG system is formulated as a multivariable feedback system and a new performance robustness criterion for performance evaluation is provided. In Ref. [27], a tuned damping controller is designed and its impact on different operating conditions of a DFIG-based generator system is investigated. However, limited research exists on the small-signal modeling and analysis of PMSG-based WTGS. Although the small-signal stability of PMSG-based WTGS is assessed in Refs. [28,29], they do not consider the interactions between WTG mechanical system, electrical and control systems. Small signal stability is also analyzed in Refs. [30–32] but they either modeled drive train as a one-mass mechanical subsystem or only considered one operating point. In addition, in aforementioned work, not all existing oscillation modes for WTGs connected to the power grid are classified and analyzed in details.

This paper focuses on studying the oscillation modes of direct-driven PMSGs connected to the power grid by developing small-signal models of PMSG-based WTGS with different converter topologies. The mechanical system of a PMSG-based WTGS, including wind turbine blade, low-speed shaft, and generator rotor, is lumped into a two-mass shaft model. Small signal models are then developed for three types of converters, PMSG and transmission lines (including transformer). In this way, the PMSG-based WTGS are modeled with several independent modules. Finally, eigenvalue analysis is conducted to study the oscillation modes of different types of PMSGs connected to the power grid. The results obtained from eigenvalue analysis based on the unified small-signal models are also demonstrated through time-domain simulation.

The main contributions of this paper are: 1) determining all oscillation modes for PMSG-based WTGS connected to power grid, including oscillation frequency and damping ratio of each mode; 2) categorizing all oscillation modes into different categories and conducting participation factor analysis to find the relative participation of system variables in each oscillation mode; 3) evaluating the influence of various initial operating states on the oscillation frequencies and damping ratios of all oscillation modes.

The remaining parts of this paper are organized as follows: Section 2 presents the topologies of different types of PMSGs. The small-signal models of a wind turbine, PMSG and transmission line are developed in Section 3. In Section 4, the small signal models of the three types of converters are deduced in detail. Unified small signal models are established for PMSGs in Section 5. In Section 6, modal analysis is conducted to find all oscillation modes and the influence of initial values on the oscillation modes is also studied for the three types of PMSGs. Conclusions are drawn in Section 7.

2. Background theory

2.1. Small signal stability

The behavior of power system can be described by the following vector-matrix equation: $\frac{d}{dt}\mathbf{X} = \mathbf{F}(\mathbf{X}, \mathbf{u}, t)$, where $\mathbf{X}$ represents state variables and $\mathbf{u}$ represents input variables. At a steady state operation point, it can be written as: $\frac{d}{dt}\mathbf{X}_{\text{point}} = \mathbf{F}(\mathbf{X}_{\text{point}}, \mathbf{u}_{\text{point}}, t)$. Subscript ‘point’ represents steady state operation point. Then it can be Taylor expanded and then linearized to: $\Delta \frac{d}{dt}\mathbf{X} = \mathbf{A}\Delta\mathbf{X} + \mathbf{B}\Delta\mathbf{u}$, where $\Delta\mathbf{X} = \mathbf{X} - \mathbf{X}_{\text{point}}$, $\Delta\mathbf{u} = \mathbf{u} - \mathbf{u}_{\text{point}}$.

$$\mathbf{A} = \begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \frac{\partial F_1}{\partial x_2} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \frac{\partial F_2}{\partial x_1} & \frac{\partial F_2}{\partial x_2} & \cdots & \frac{\partial F_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_n}{\partial x_1} & \frac{\partial F_n}{\partial x_2} & \cdots & \frac{\partial F_n}{\partial x_n} \end{bmatrix}_{\substack{\mathbf{X}=\mathbf{X}_{\text{point}} \\ \mathbf{u}=\mathbf{u}_{\text{point}}}}, \quad \mathbf{B} = \begin{bmatrix} \frac{\partial F_1}{\partial u_1} & \frac{\partial F_1}{\partial u_2} & \cdots & \frac{\partial F_1}{\partial u_n} \\ \frac{\partial F_2}{\partial u_1} & \frac{\partial F_2}{\partial u_2} & \cdots & \frac{\partial F_2}{\partial u_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_n}{\partial u_1} & \frac{\partial F_n}{\partial u_2} & \cdots & \frac{\partial F_n}{\partial u_n} \end{bmatrix}_{\substack{\mathbf{X}=\mathbf{X}_{\text{point}} \\ \mathbf{u}=\mathbf{u}_{\text{point}}}}.$$

The eigenvalues of the state matrix $\mathbf{A}$ reveal the time domain response of the system to small disturbances, showing information about the small signal stability characteristics. For a complex conjugate pair of eigenvalues $\lambda_i = \sigma_i \pm j\omega_i$, the frequency of the oscillation is $f_i = \frac{\omega_i}{2\pi}$ and the damping ratio is $\xi_i = \frac{-\sigma_i}{\sqrt{\sigma_i^2 + \omega_i^2}}$. For a real eigenvalue, the corresponding time domain response does not change periodicity.

Identifying the association between the k th state variable and the i th oscillation mode, an element $P_{ki} = \phi_{ki}\psi_{ik}$ called the participation factor is proposed, where ϕ_{ki}, ψ_{ik} are the k th entry of the left and right eigenvectors of the i th oscillation mode λ_i [33].

2.2. Topologies of different types of PMSGs

Two types of power converter techniques are usually adopted for PMSGs: back-to-back PWM converter system and DC boost converter system. In the PMSG-based WTGS with back-to-back PWM full rate converter, decoupled d-q vector control and power signal feedback control are the two commercial control schemes we place attention on for generator side rectifier (GSR) control. There are many other control schemes which are mainly studied in theory without large-scale commercial application or cannot be modeled with perfect precision from wind turbine makers. Those types’ models are not provided due to study limitation [34–37]. Given the high electricity losses of power electronics and the low efficiency of a back-to-back system, the DC boost converter topology is also widely used [38]. In this section, the two types of power converter techniques are discussed in detail.

2.2.1. PMSG with back-to-back PWM converter system

PMSG with a full-scale back-to-back PWM converter system is widely used for its high fault ride through and grid voltage support ability [39]. A PMSG-based wind turbine with a back-to-back PWM converter system is made up of a wind turbine, a PMSG and two voltage source converters (VSCs), shown in Fig. 1. Decoupled d-q vector control and power signal feedback control schemes are the two commercial control schemes we place attention on for the generator side rectifier (GSR), as shown in Fig. 1 (a) and (b) respectively.

2.2.1.1. Decoupled d-q vector control system. In Fig. 1 (a), the GSR is controlled through a decoupled d-q vector control approach. The d-axis current i_{ds} is controlled to be zero to limit the output reactive

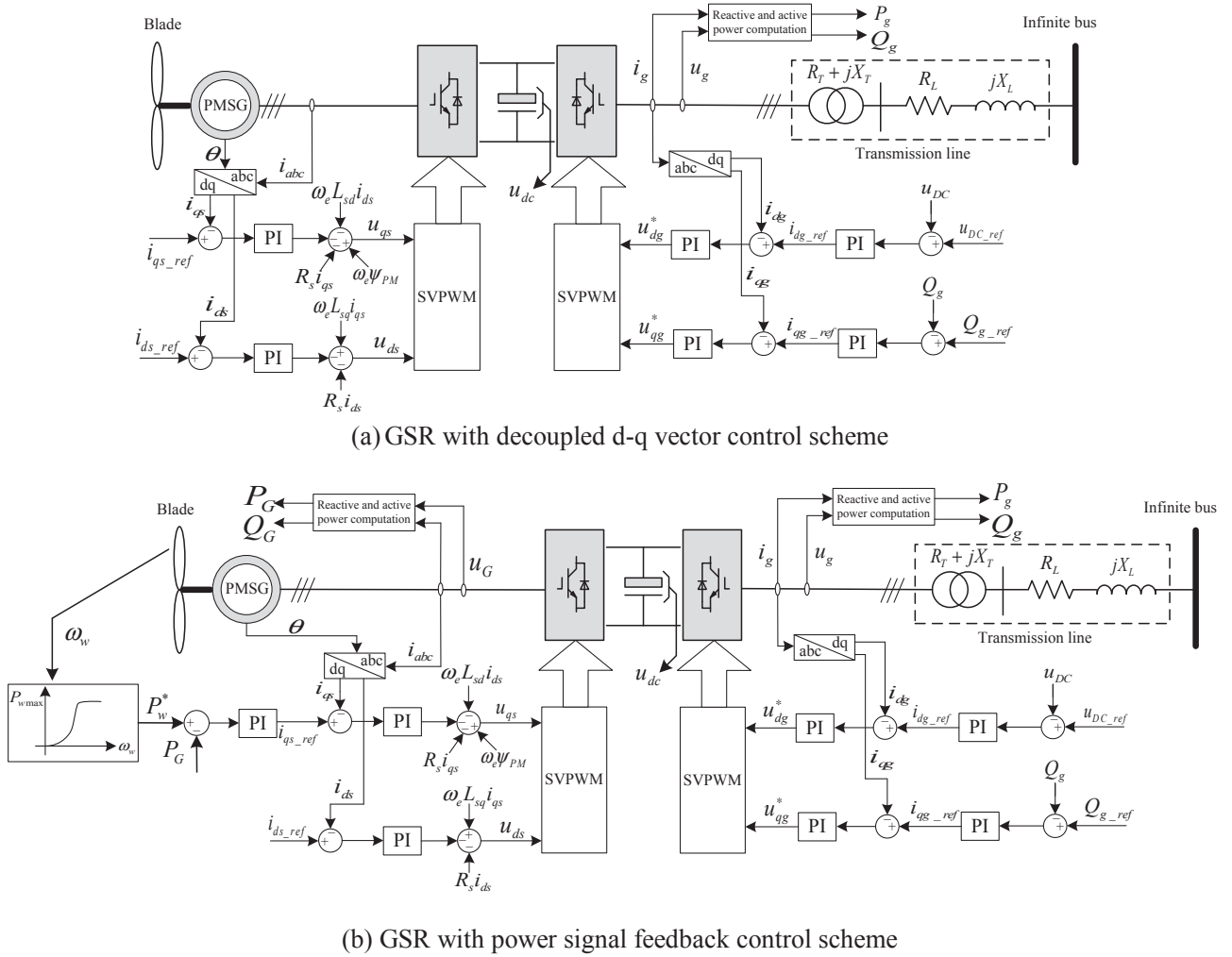


Fig. 1. PMSG with back-to-back PWM converter system.

power of the generator. The q-axis current i_{qs} is controlled to track the electrical torque of the generator to obtain the maximum active power. The voltage source inverter (VSI) is controlled to maintain the dc-link voltage a constant and achieve zero reactive power.

2.2.1.2. Power signal feedback control system. Fig. 1 (b) is almost the same with Fig. 1 (a). The difference between the two control schemes is that an MPPT control is implemented on the GSR. For each turbine rotating speed ω_w , there exists a corresponding maximum power point, as the $P_{w\max} - \omega_w$ curve shown in Fig. 1 (b). The q-axis reference current i_{qs_ref} is obtained from $P_{w\max}$ through a PI controller.

2.2.2. PMSG with a DC boost converter system

The DC boost converter topology is also widely applied for PMSGs, which is made up of a diode rectifier, a boost-up DC/DC converter and a VSI, shown in Fig. 2. The control scheme of the DC/DC chopper is composed of two control loops. The outer loop is used to control the output power of PMSG and the inner loop to control the input current of chopper through adjusting the duty ratio. The control scheme of the VSI is the same with that of Fig. 1 (a) and (b). Compared with the back-to-back PWM converter system, the DC boost system allows for specific harmonic elimination and gives high efficiency and flexible control [38].

3. Model of grid-connected PMSGs

As seen in Figs. 1 and 2, the PMSG system has four parts: a wind turbine, a PMSG, a converter system and a transmission line. The PMSG is directly driven by a wind turbine and the output power is delivered to the grid through a converter system and transmission line (including the transformer and cable). Except for converter systems, other parts of the three types of PMSGs are the same. The small signal models of identical parts (wind turbine, PMSG and transmission line) are developed in this section.

3.1. Wind turbine model

For a direct-driven PMSG-based wind turbine, the mechanical system is composed of a wind turbine blade, a low-speed shaft, and a generator rotor. As for WTGS, rotating shaft systems can be divided into masses to describe the physical characteristics of the real system and accurate results can be obtained by increasing the number of masses and freedom degrees [40–42]. In Ref. [43], the transient stability of a WTGS connected to the power systems is analyzed using a six-mass, a three-mass, and a two-mass drive-train model respectively. It has been shown that two-mass shaft model is sufficient, with reasonable accuracy, for the transient stability analysis. In Ref. [44], the dynamic performance of wind

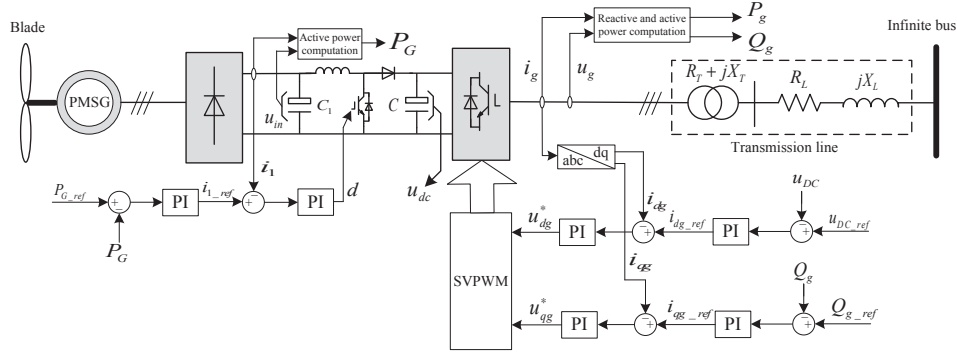


Fig. 2. PMSG with DC boost converter system.

turbines is assessed. The paper concluded that a three-mass model taking shaft and blade flexibilities into account could be reduced to an effective two-mass model. A two-mass model for the mechanical block of PMSG is thus adopted, shown in Fig. 3. In Fig. 3 (b), the first lumped mass represents heavy and big blades, and the second lumped mass represents the low-speed shaft and the rotor of the PMSG. The nomenclature of variables in Fig. 3 is as follows: i) T_w is the aerodynamic torque from the wind turbine; ii) T_e is the electromechanical torque from the generator; iii) J_1 and J_2 are the moments of inertia of blades and generator rotor respectively; iv) D_1 and D_2 are the damping of blades and generator rotor respectively; v) K_{12} and D_{12} are stiffness and damping coefficient between blades and generator rotor respectively.

Based on Hooke's law and Newton's second law, the two-mass shaft model of a PMSG wind turbine can be expressed by the following linearized equation:

$$\begin{cases} \Delta \dot{\theta}_1 = \Delta \omega_1 \\ \Delta \dot{\theta}_2 = \Delta \omega_2 \\ J_1 \Delta \dot{\omega}_1 = \Delta T_w - D_1 \Delta \omega_1 - D_{12} (\Delta \omega_1 - \Delta \omega_2) - K_{12} (\Delta \theta_1 - \Delta \theta_2) \\ J_2 \Delta \dot{\omega}_2 = \Delta T_e - D_2 \Delta \omega_2 - D_{12} (\Delta \omega_2 - \Delta \omega_1) - K_{12} (\Delta \theta_2 - \Delta \theta_1) \end{cases} \quad (1)$$

where " Δ " denotes a small deviation; ω_1 and θ_1 are angular velocity and twist angle of the wind turbine respectively; ω_2 and θ_2 are angular velocity and twist angle of the generator rotor respectively.

The small signal standard state-space equations of (1) can be obtained as follows:

$$\begin{cases} \Delta \dot{\mathbf{X}}_M = \mathbf{A}_M \Delta \mathbf{X}_M + \mathbf{B}_M \Delta \mathbf{u}_M \\ \Delta \mathbf{Y}_M = \mathbf{C}_M \Delta \mathbf{X}_M + \mathbf{D}_M \Delta \mathbf{u}_M \end{cases} \quad (2)$$

where subscript M refers to the mechanical system.

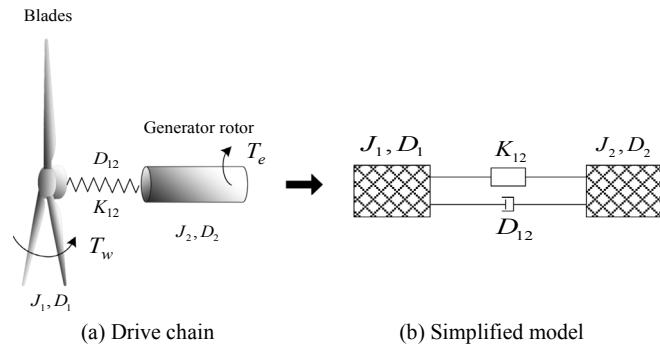


Fig. 3. The two-mass Shaft model of a PMSG wind turbine.

Other parameters in (2) are

$$\begin{aligned} \Delta \mathbf{X}_M &= [\Delta \theta_1 \quad \Delta \theta_2 \quad \Delta \omega_1 \quad \Delta \omega_2]^T; & \Delta \mathbf{Y}_M &= [\Delta \theta_2 \quad \Delta \omega_2]^T; \\ \Delta \mathbf{u}_M &= [\Delta T_w \quad \Delta T_e]^T; \\ \mathbf{A}_M &= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{K_{12}}{J_1} & \frac{K_{12}}{J_1} & \frac{D_1 + D_{12}}{J_1} & \frac{D_{12}}{J_1} \\ \frac{K_{12}}{J_2} & \frac{K_{12}}{J_2} & \frac{D_{12}}{J_2} & \frac{D_2 + D_{12}}{J_2} \end{bmatrix}; \\ \mathbf{B}_M &= \begin{bmatrix} 0 & 0 & \frac{1}{J_1} & 0 \\ 0 & 0 & 0 & \frac{1}{J_2} \end{bmatrix}^T; & \mathbf{C}_M &= \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}; & \mathbf{D}_M &= \mathbf{0}_{2 \times 2}. \end{aligned}$$

3.2. Model of PMSG

A PMSG is usually modeled in the dq-reference frame, where d-axis is magnetically centered in the center of the north pole [45]. The voltage equation for the PMSG is expressed as

$$\begin{cases} u_{ds} = -R_s i_{ds} - \omega_r \psi_{qs} + \dot{\psi}_{ds} \\ u_{qs} = -R_s i_{qs} + \omega_r \psi_{ds} + \dot{\psi}_{qs} \end{cases} \quad (3)$$

where ψ_{ds} and ψ_{qs} are the flux linkages of stator windings in the d- and q-axis respectively; u_{ds} and u_{qs} are the voltage of stator windings against the d- and q-axis respectively; i_{ds} and i_{qs} are the currents of stator windings against the d- and q-axis respectively.

The flux linkage equation can be written as:

$$\begin{cases} \psi_{ds} = -L_{sd} i_{ds} + \psi_{rd} \\ \psi_{qs} = -L_{sq} i_{qs} + \psi_{rq} \\ \psi_{rd} = \psi_{PM} \\ \psi_{rq} = 0 \end{cases} \quad (4)$$

where L_{sd} and L_{sq} are the inductance of stator windings against the d- and q-axis respectively, and ψ_{rq} are the flux linkage of rotor windings against the d- and q-axis respectively; ψ_{PM} is the amplitude of flux linkage of the rotating magnet.

The electromagnetic torque is

$$T_e = \frac{3}{2} n_p (\psi_{ds} i_{qs} - \psi_{qs} i_{ds}) \quad (5)$$

For a non-salient pole PMSG adopted in this paper, the relationship between d- and q-axis inductance is

$$L_{sd} = L_{sq} \quad (6)$$

Based on (3)–(6), the small signal standard state space equations for a PMSG with fluxes as state variables are:

$$\begin{cases} \Delta \dot{\mathbf{X}}_G = \mathbf{A}_G \Delta \mathbf{X}_G + \mathbf{B}_G \Delta \mathbf{u}_G \\ \Delta \mathbf{Y}_G = \mathbf{C}_G \Delta \mathbf{X}_G + \mathbf{D}_G \Delta \mathbf{u}_G \end{cases} \quad (7)$$

where subscript G refers to PMSG.

Other parameters in (7) are represented as follows

$$\begin{aligned} \Delta \mathbf{X}_G &= [\Delta \psi_{ds} \quad \Delta \psi_{qs}]^T; \Delta \mathbf{Y}_G = [\Delta i_{ds} \quad \Delta i_{qs} \quad \Delta T_e]^T; \\ \Delta \mathbf{u}_G &= [\Delta u_{ds} \quad \Delta u_{qs} \quad \Delta \omega_r]^T; \\ \mathbf{A}_G &= \begin{bmatrix} -\frac{R_s}{L_{sd}} & \omega_b \\ -\omega_b & -\frac{R_s}{L_{sq}} \end{bmatrix}; \mathbf{B}_G = \begin{bmatrix} 1 & 0 & \psi_{qs0} \\ 0 & 1 & \psi_{ds0} \end{bmatrix}; \\ \mathbf{C}_G &= \begin{bmatrix} -\frac{1}{L_{sd}} & 0 & 0 \\ 0 & -\frac{1}{L_{sq}} & -\frac{3n_p \psi_{PM}}{2L_{sq}} \end{bmatrix}^T; \mathbf{D}_G = \mathbf{0}_{3 \times 3}. \end{aligned}$$

where ω_b is the base angular velocity of the system; subscript "0" denotes the initial values of variables in steady state.

3.3. Model of the transmission line

The transmission line is shielded cable, which is usually equipped with a series compensation capacitor to improve the power transfer capacity. For the convenience of study, the transformer and the transmission line are modeled as an equivalent RLC (resistor, inductor, and capacitor) line. The standard state space equations are

$$\begin{cases} \Delta \dot{\mathbf{X}}_{RLC} = \mathbf{A}_{RLC} \Delta \mathbf{X}_{RLC} + \mathbf{B}_{RLC} \Delta \mathbf{u}_{RLC} \\ \Delta \mathbf{Y}_{RLC} = \mathbf{C}_{RLC} \Delta \mathbf{X}_{RLC} + \mathbf{D}_{RLC} \Delta \mathbf{u}_{RLC} \end{cases} \quad (8)$$

where subscript RLC refers to the transmission line.

Other parameters in (8) are represented as follows

$$\begin{aligned} \Delta \mathbf{X}_{RLC} &= [\Delta i_x \quad \Delta i_y \quad \Delta u_{cx} \quad \Delta u_{cy}]^T; \Delta \mathbf{Y}_{RLC} = [\Delta i_x \quad \Delta i_y]^T; \\ \Delta \mathbf{u}_{RLC} &= [\Delta u_{x1} \quad \Delta u_{y1} \quad \Delta u_{x2} \quad \Delta u_{y2}]^T; \\ \mathbf{A}_{RLC} &= \begin{bmatrix} -\omega_b R_{TL}/X_{TL} & \omega_b & -\omega_b/X_{TL} & 0 \\ -\omega_b & -\omega_b R_{TL}/X_{TL} & 0 & -\omega_b/X_{TL} \\ -\omega_b X_C & 0 & 0 & \omega_b \\ 0 & -\omega_b X_C & -\omega_b & 0 \end{bmatrix}; \\ \mathbf{B}_{RLC} &= \begin{bmatrix} \omega_b/X_{TL} & 0 & -\omega_b/X_{TL} & 0 \\ 0 & \omega_b/X_{TL} & 0 & -\omega_b/X_{TL} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}; \\ \mathbf{C}_{RLC} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}; \mathbf{D}_{RLC} = \mathbf{0}_{2 \times 4}. \end{aligned}$$

where $R_{TL} = R_T + R_L$, $X_{TL} = X_T + X_L$, R_T and X_T are resistance and reactance of the transformer respectively; R_L and X_L are resistance and reactance of the cable respectively; X_C is the reactance of the series compensation capacitor. Δi and Δu_c denote a small deviation of the current of the inductor and the voltage of the series compensation capacitor respectively. u_{x1} and u_{y1} are the entrance voltage of the transmission line against the x- and y-axis respectively. u_{x2} and u_{y2} are the outlet voltage of the transmission line against the x- and y-axis respectively.

4. Small-signal model of converter systems

4.1. Model of the GSR with decoupled d-q vector control

The schematic diagram of the GSR is shown in Fig. 1 (a) and the equations of the controller are shown in (9).

$$\begin{cases} i_{ds-ref} = 0 \\ i_{qs-ref} = \frac{2T_e}{3n_p \psi_{PM}} \\ \dot{x}_1 = i_{qs-ref} - i_{qs} \\ \dot{x}_2 = i_{ds-ref} - i_{ds} \\ u_{qs} = \omega_e \psi_{PM} - \omega_e L_{sd} i_{ds} - R_s i_{qs} - K_{p1} \dot{x}_1 - K_{i1} x_1 \\ u_{ds} = \omega_e L_{sq} i_{qs} - R_s i_{ds} - K_{p2} \dot{x}_2 - K_{i2} x_2 \end{cases} \quad (9)$$

where, x_1 and x_2 are intermediate state variables.

According to equation (9), the standard state space equations after linearising are

$$\begin{cases} \Delta \dot{\mathbf{X}}_{GSR} = \mathbf{A}_{GSR} \Delta \mathbf{X}_{GSR} + \mathbf{B}_{GSR} \Delta \mathbf{u}_{GSR} \\ \Delta \mathbf{Y}_{GSR} = \mathbf{C}_{GSR} \Delta \mathbf{X}_{GSR} + \mathbf{D}_{GSR} \Delta \mathbf{u}_{GSR} \end{cases} \quad (10)$$

where subscript GSR refers to the generator side rectifier.

Other parameters in (10) are represented as follows

$$\begin{aligned} \Delta \mathbf{X}_{GSR} &= [\Delta x_1 \quad \Delta x_2]^T; \Delta \mathbf{Y}_{GSR} = [\Delta u_{ds} \quad \Delta u_{qs}]^T; \\ \Delta \mathbf{u}_{GSR} &= [\Delta i_{ds} \quad \Delta i_{qs} \quad \Delta T_e \quad \Delta \omega_e]^T; \mathbf{A}_{GSR} = \mathbf{0}_{2 \times 2}; \\ \mathbf{B}_{GSR} &= \begin{bmatrix} 0 & -1 & \frac{2}{3n_p \psi_{PM}} & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix}; \mathbf{C}_{GSR} = \begin{bmatrix} 0 & -K_{i2} \\ -K_{i1} & 0 \end{bmatrix}; \\ \mathbf{D}_{GSR} &= \begin{bmatrix} K_{p2} - R_s & \omega_b L_{sq} & 0 & L_{sq} i_{ds0} \\ -\omega_b L_{sd} & K_{p1} - R_s & -\frac{2K_{p1}}{3n_p \psi_{PM}} & \psi_{PM} \end{bmatrix}. \end{aligned}$$

4.2. Model of the GSR with power signal feedback control

The schematic diagram of the GSR is shown in Fig. 1 (b) and the equations of the controller are shown in (11).

$$\begin{cases} i_{ds-ref} = 0 \\ \dot{x}_1 = P_{w-ref} - P_G \\ \dot{x}_2 = i_{qs-ref} - i_{qs} \\ \dot{x}_3 = i_{ds-ref} - i_{ds} \\ i_{qs-ref} = K_{pm} \dot{x}_1 + K_{im} x_1 \\ u_{ds} = \omega_e L_{sq} i_{qs} - R_s i_{ds} - K_{p2} \dot{x}_3 - K_{i2} x_3 \\ u_{qs} = \omega_e \psi_{PM} - \omega_e L_{sd} i_{ds} - R_s i_{qs} - K_{p1} \dot{x}_2 - K_{i1} x_2 \end{cases} \quad (11)$$

where, x_1 , x_2 and x_3 are intermediate state variables.

After linearising equation (11), the small signal model can be expressed as

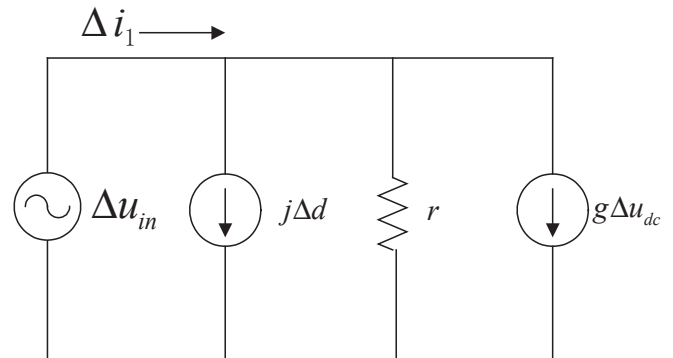


Fig. 4. Input-stage small-signal equivalent circuit of a DC/DC chopper.

$$\begin{cases} \Delta \dot{\mathbf{X}}_{GSR} = \mathbf{A}_{GSR} \Delta \mathbf{X}_{GSR} + \mathbf{B}_{GSR} \Delta \mathbf{u}_{GSR} \\ \Delta \mathbf{Y}_{GSR} = \mathbf{C}_{GSR} \Delta \mathbf{X}_{GSR} + \mathbf{D}_{GSR} \Delta \mathbf{u}_{GSR} \end{cases} \quad (12)$$

where subscript GSR refers to the generator side rectifier.

Other parameters in (12) are represented as follows

$$\begin{aligned} \Delta \mathbf{X}_{GSR} &= [\Delta x_1 \quad \Delta x_2 \quad \Delta x_3]^T; \Delta \mathbf{Y}_{GSR} = [\Delta u_{ds} \quad \Delta u_{qs}]^T; \\ \Delta \mathbf{u}_{GSR} &= [\Delta u_{ds} \quad \Delta u_{qs} \quad \Delta i_{ds} \quad \Delta i_{qs} \quad \Delta P_{w_ref} \quad \Delta \omega_e]^T; \\ \mathbf{A}_{GSR} &= \begin{bmatrix} 0 & 0 & 0 \\ K_{im} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}; \\ \mathbf{B}_{GSR} &= \begin{bmatrix} -i_{ds0} & -i_{qs0} & -u_{ds0} & -u_{qs0} & 1 & 0 \\ -K_{pm}i_{ds0} & -K_{pm}i_{qs0} & -K_{pm}u_{ds0} & -K_{pm}u_{qs0} & K_{pm} & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \end{bmatrix}; \\ \mathbf{C}_{GSR} &= \mathbf{0}_{2 \times 3}; \mathbf{D}_{GSR} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}. \end{aligned}$$

4.3. Model of the DC/DC chopper

The small-signal model of the boost-up DC/DC chopper is designed based on the equivalent small-signal circuit in Ref. [46], shown in Fig. 4. Δd is the small deviation of duty ratio and the model can be expressed as:

$$\begin{cases} \Delta i_1 = j\Delta d + \frac{\Delta u_{in}}{r} + g\Delta u_{dc} \\ u_{in} = \frac{3\sqrt{2}}{\pi} u_{L-L} \end{cases} \quad (13)$$

where u_{L-L} is the output line-to-line voltage of the generator.

j , r and g can be obtained from (14).

$$\begin{cases} j = \frac{0.96DV_{in}V_{dc}T_s}{L(V_{dc} - 1.67V_{in})} \\ g = \left(\frac{0.48V_{in}D^2T_s}{L} - I_1 \right) \frac{1}{V_{dc} - 0.92V_{in}} \\ I_1 = \frac{V_{in}V_{dc}D^2T_s}{L} \frac{0.48}{V_{dc} - 0.92V_{in}} \\ r = -\frac{V_{dc} - 0.92V_{in}}{\frac{0.48V_{dc}D^2T_s}{L} + 0.92I_1} \\ L = \frac{1.46V_{dc}^2D^2T_s}{2M(M - 1.67)P} \end{cases} \quad (14)$$

where V_{in} and V_{dc} are the steady-state voltage of C_1 and C respectively; D is the steady-state duty ratio; T_s is the switching period; P is the rated output power; $M = \sqrt{3}/(1 - D)$ [46].

The control scheme of DC/DC chopper is shown in Fig. 2. By linearising the control scheme, the standard state space of the DC chopper with x_1 and x_2 as intermediate variables is

$$\begin{cases} \Delta \dot{\mathbf{X}}_S = \mathbf{A}_S \Delta \mathbf{X}_S + \mathbf{B}_S \Delta \mathbf{u}_S \\ \Delta \mathbf{Y}_S = \mathbf{C}_S \Delta \mathbf{X}_S + \mathbf{D}_S \Delta \mathbf{u}_S \end{cases} \quad (15)$$

where subscript S refers to DC/DC chopper.

Other parameters in (15) are represented as follows

$$\begin{aligned} \Delta \mathbf{X}_S &= [\Delta x_1 \quad \Delta x_2]^T; \Delta \mathbf{Y}_S = [\Delta d]; \Delta \mathbf{u}_S = [\Delta P_{G.ref} \quad \Delta u_{ds} \quad \Delta u_{dc}]^T; \\ \mathbf{A}_S &= \begin{bmatrix} 0 & j_1[(b+1)K_{i2}e - K_{i2}u_{in0}] \\ \frac{K_{i1}}{a} & -\frac{(b+1)j_1K_{i2}}{a} \end{bmatrix}; \\ \mathbf{B}_S &= \begin{bmatrix} \frac{K_{p1}e+1}{f} & \frac{9[(b+c+1)e - i_{10} - u_{in0}]}{\sqrt{3}\pi f} & \frac{g_1[(b+1)e - u_{in0}]}{f} \\ \frac{K_{p1}}{a} & \frac{9(b+c+1)}{\sqrt{3}\pi a} & -\frac{(b+1)g_1}{a} \end{bmatrix}; \\ \mathbf{C}_S &= \begin{bmatrix} \frac{K_{p2}K_{i1}}{a} & K_{i2} - \frac{(b+1)j_1K_{p2}K_{i2}}{a} \end{bmatrix}; \\ \mathbf{D}_S &= \begin{bmatrix} \frac{K_{p2}K_{p1}}{a} & -\frac{9(b+c+1)K_{p2}}{\sqrt{3}\pi a} & -\frac{K_{p2}(b+1)g_1}{a} \end{bmatrix}. \end{aligned}$$

$$a = 1 + k_{p2}(b+1)j; b = k_{p1}u_{in0}; c = k_{p1}i_{10}; e = \frac{k_{p2}u_{in0}j}{a}; f = 1 + k_{i1}e.$$

4.4. Model of the DC-link

The power balance equation of the DC-link is expressed as

$$Cu_{dc}\dot{u}_{dc} = u_{dg}i_{dg} + u_{qg}i_{qg} - (u_{ds}i_{ds} + u_{qs}i_{qs}) \quad (16)$$

The dynamic small-signal model of the DC-link can be deduced by linearising (16), shown as

$$\begin{cases} \Delta \dot{\mathbf{X}}_{dc} = \mathbf{A}_{dc} \Delta \mathbf{X}_{dc} + \mathbf{B}_{dc} \Delta \mathbf{u}_{dc} \\ \Delta \mathbf{Y}_{dc} = \mathbf{C}_{dc} \Delta \mathbf{X}_{dc} + \mathbf{D}_{dc} \Delta \mathbf{u}_{dc} \end{cases} \quad (17)$$

where, subscript dc refers to the DC-link.

Other parameters in (17) are represented as follows

$$\begin{aligned} \Delta \mathbf{X}_{dc} &= \Delta \mathbf{Y}_{dc} = [\Delta u_{dc}]^T; \\ \Delta \mathbf{u}_{dc} &= [\Delta u_{qg} \quad \Delta u_{dg} \quad \Delta i_{qg} \quad \Delta i_{dg} \quad \Delta u_{qs} \quad \Delta u_{ds} \quad \Delta i_{qs} \quad \Delta i_{ds}]^T; \\ \mathbf{A}_{dc} &= \mathbf{0}_{1 \times 1}; \mathbf{B}_{dc} = [i_{qg0} \quad i_{dg0} \quad u_{qg0} \quad u_{dg0} \quad -i_{qs0} \quad -i_{ds0} \quad -u_{qs0} \quad -u_{ds0}]; \\ \mathbf{C}_{dc} &= \mathbf{I}_{1 \times 1}; \mathbf{D}_{dc} = \mathbf{0}_{1 \times 8}. \end{aligned}$$

4.5. Model of the voltage source inverter

The control schemes of the VSI in Figs. 1 and 2 are the same. Through linearising the control scheme, the standard state-space equations are

$$\begin{cases} \Delta \dot{\mathbf{X}}_{VSI} = \mathbf{A}_{VSI} \Delta \mathbf{X}_{VSI} + \mathbf{B}_{VSI} \Delta \mathbf{u}_{VSI} \\ \Delta \mathbf{Y}_{VSI} = \mathbf{C}_{VSI} \Delta \mathbf{X}_{VSI} + \mathbf{D}_{VSI} \Delta \mathbf{u}_{VSI} \end{cases} \quad (18)$$

where subscript VSI refers to the voltage source inverter.

Other parameters in (18) are represented as follows

$$\begin{aligned} \Delta \mathbf{X}_{VSI} &= [\Delta x_3 \quad \Delta x_4 \quad \Delta x_5 \quad \Delta x_6]^T; \Delta \mathbf{Y}_{VSI} = [\Delta u_{gd} \quad \Delta u_{gq}]^T; \Delta \mathbf{u}_{VSI} = [\Delta u_{DC.ref} \quad \Delta Q_{g.ref} \quad \Delta u_{DC} \quad \Delta u_{gd} \quad \Delta u_{gq} \quad \Delta i_{gd} \quad \Delta i_{gq}]^T; \\ \mathbf{A}_{VSI} &= \begin{bmatrix} 0 & 0 & 0 & 0 \\ K_{i3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & K_{i5} & 0 \end{bmatrix}; \mathbf{B}_{VSI} = \begin{bmatrix} 1 & 0 & -1 & 0 & 0 & 0 & 0 \\ K_{p3} & 0 & -K_{p3} & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & i_{gq0} & -i_{gd0} & -u_{gq0} & u_{gd0} \\ 0 & K_{p5} & 0 & K_{p5}i_{gq0} & -K_{p5}i_{gd0} & -K_{p5}u_{gq0} & K_{p5}u_{gd0} - 1 \end{bmatrix}; \mathbf{C}_{VSI} = \begin{bmatrix} K_{i3}K_{p4} & K_{i4} & 0 & 0 \\ 0 & 0 & K_{i5}K_{p5} & K_{i6} \end{bmatrix}; \\ \mathbf{D}_{VSI} &= \begin{bmatrix} K_{p3}K_{p4} & 0 & -K_{p3}K_{p4} & 0 & 0 & -K_{p3} & 0 \\ 0 & K_{p5}K_{p6} & 0 & K_{p5}K_{p6}i_{gq0} & -K_{p5}K_{p6}i_{gd0} & -K_{p5}K_{p6}u_{gq0} & (K_{p5}u_{gd0} - 1)K_{p6} \end{bmatrix}. \end{aligned}$$

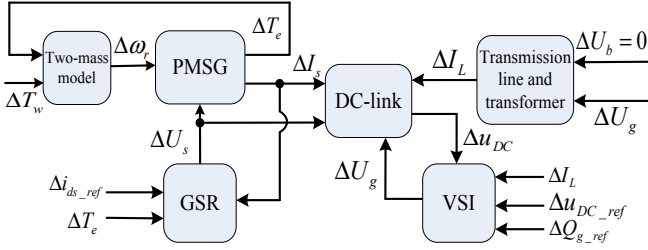


Fig. 5. Block diagram of small-signal model of PMSG-I.

5. Unified small-signal models of PMSGs

To simplify the research, back-to-back PWM based PMSG with decoupled d-q vector controlled GSR is defined as PMSG-I, with power signal feedback controlled GSR defined as PMSG-II. DC boost based PMSG is then defined as PMSG-III. In this section, the small signal models of the three PMSGs are developed, which are integrated with several modules, i.e. two-mass shaft system, PMSG, converter and transmission lines.

5.1. Unified small-signal model of PMSG-I

Based on the small-signal models of each component, the block diagram of a small-signal model of PMSG-I is illustrated in Fig. 5, showing the small signal stability for PMSG-I system.

In Fig. 5, ΔT_w and ΔT_e are the input aerodynamic torque and damping torque of the two-mass model respectively. The output is the generator rotor speed $\Delta \omega_r$, which is also the input of PMSG. Expect for ΔT_e , another output of PMSG is stator current ΔI_s . The input of GSR includes three parts: d-axis reference current Δi_{ds_ref} , ΔT_e and ΔI_s . The output of GSR is the generator voltage ΔU_s , also the input of PMSG. As for the VSI, the input consists of transmission line current ΔI_L , the reference voltage of DC-link Δu_{DC_ref} , reference reactive power ΔQ_{g_ref} , and DC-link voltage Δu_{DC} . The output of VSI is the grid-side voltage ΔU_g . To achieve power balance of the DC-link, the input of DC-link is ΔU_s , ΔI_s , ΔU_g and ΔI_L , and the output is Δu_{DC} . For the integrated model of the transmission line and transformer, the input is the infinite bus voltage ΔU_b and ΔU_g , and the output is ΔI_L .

Based on equations (2), (7), (8), (10), (17) and (18), the unified small signal model of PMSG-I can be obtained according to Fig. 5, shown as

$$\Delta \dot{\mathbf{X}}_{PMSG-I} = \mathbf{A}_{PMSG-I} \Delta \mathbf{X}_{PMSG-I} + \mathbf{B}_{PMSG-I} \Delta \mathbf{u}_{PMSG-I} \quad (19)$$

where, $\Delta \mathbf{X}_{PMSG-I} = [\Delta \mathbf{X}_M^T \ \Delta \mathbf{X}_G^T \ \Delta \mathbf{X}_{GSR}^T \ \Delta \mathbf{X}_{DC}^T \ \Delta \mathbf{X}_{VSI}^T \ \Delta \mathbf{X}_{RLC}^T]^T$;

$$\Delta \mathbf{u}_{PMSG-I} = [\Delta T_w \ \Delta i_{ds_ref} \ \Delta u_{DC_ref} \ \Delta Q_{g_ref} \ \Delta U_b]^T;$$

$$\mathbf{A}_{PMSG-I} = \begin{bmatrix} \mathbf{A}_M & \mathbf{A}_{G,M} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{A}_{M,G} & \mathbf{A}_G & \mathbf{A}_{GSR,G} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{G,GSR} & \mathbf{A}_{GSR} & \mathbf{A}_{DC,GSR} & \mathbf{0} & \mathbf{0} \\ \mathbf{A}_{M,DC} & \mathbf{A}_{G,DC} & \mathbf{A}_{GSR,DC} & \mathbf{A}'_{DC} & \mathbf{A}_{VSI,DC} & \mathbf{A}_{RLC,DC} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{A}_{DC,VSI} & \mathbf{A}_{VSI} & \mathbf{A}_{RLC,VSI} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{A}_{DC,RLC} & \mathbf{A}_{VSI,RLC} & \mathbf{A}'_{RLC} \end{bmatrix}.$$

Because $\mathbf{B}_{PMSG-I}$ is irrelevant to the eigenvalues of the small signal model, it is not presented here. The unknown submatrices in $\mathbf{A}_{PMSG-I}$ are shown in Appendix A.

5.2. Unified small-signal model of PMSG-II

Similar to PMSG-I, the block diagram of a small-signal model of PMSG-II is illustrated in Fig. 6.

In Fig. 6, only the control of GSR is different from that of Fig. 5.

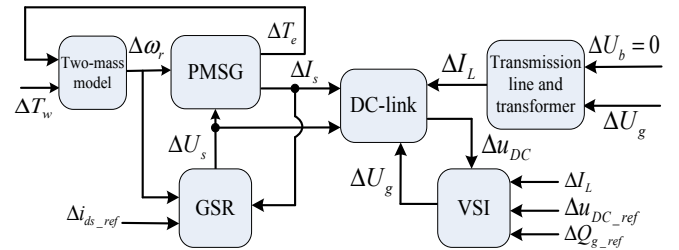


Fig. 6. Block diagram of small-signal model of PMSG-II.

For PMSG-II, one input of GSR is $\Delta \omega_r$ to track the maximum power point. Based on equations (2), (7), (8), (12), (17) and (18), the unified small signal model of PMSG-II can be obtained according to Fig. 6, shown as

$$\Delta \dot{\mathbf{X}}_{PMSG-II} = \mathbf{A}_{PMSG-II} \Delta \mathbf{X}_{PMSG-II} + \mathbf{B}_{PMSG-II} \Delta \mathbf{u}_{PMSG-II} \quad (20)$$

where,

$$\Delta \mathbf{X}_{PMSG-II} = [\Delta \mathbf{X}_M^T \ \Delta \mathbf{X}_G^T \ \Delta \mathbf{X}_{GSR}^T \ \Delta \mathbf{X}_{DC}^T \ \Delta \mathbf{X}_{VSI}^T \ \Delta \mathbf{X}_{RLC}^T]^T;$$

$$\mathbf{u}_{PMSG-II} = [\Delta T_w \ \Delta i_{ds_ref} \ \Delta u_{DC_ref} \ \Delta Q_{g_ref} \ \Delta U_b]^T;$$

$$\mathbf{A}_{PMSG-II} = \begin{bmatrix} \mathbf{A}_M & \mathbf{A}_{G,M} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{A}_{M,G} & \mathbf{A}_G & \mathbf{A}_{GSR,G} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{G,GSR} & \mathbf{A}_{GSR} & \mathbf{A}_{DC,GSR} & \mathbf{0} & \mathbf{0} \\ \mathbf{A}_{M,DC} & \mathbf{A}_{G,DC} & \mathbf{A}_{GSR,DC} & \mathbf{A}'_{DC} & \mathbf{A}_{VSI,DC} & \mathbf{A}_{RLC,DC} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{A}_{DC,VSI} & \mathbf{A}_{VSI} & \mathbf{A}_{RLC,VSI} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{A}_{DC,RLC} & \mathbf{A}_{VSI,RLC} & \mathbf{A}'_{RLC} \end{bmatrix}.$$

The unknown submatrices in $\mathbf{A}_{PMSG-II}$ are shown in Appendix B.

5.3. Unified small-signal model of PMSG-III

PMSG-III is different from the other two types of PMSGs in that the GSR is replaced with a diode rectifier and a boost-up DC/DC chopper. The block diagram of a small-signal model of PMSG-III is illustrated in Fig. 7.

In Fig. 7, reference generator power ΔP_{G_ref} and ΔI_s are the two inputs of the DC/DC chopper. The other parts are connected similarly to PMSG-I and PMSG-II. Based on equations (2), (7), (8), (15), (17) and (18), the unified small signal model of PMSG-III can be obtained according to Fig. 7, shown as

$$\Delta \dot{\mathbf{X}}_{PMSG-III} = \mathbf{A}_{PMSG-III} \Delta \mathbf{X}_{PMSG-III} + \mathbf{B}_{PMSG-III} \Delta \mathbf{u}_{PMSG-III} \quad (21)$$

where, $\Delta \mathbf{X}_{PMSG-III} = [\Delta \mathbf{X}_M^T \ \Delta \mathbf{X}_G^T \ \Delta \mathbf{X}_S^T \ \Delta \mathbf{X}_{DC}^T \ \Delta \mathbf{X}_{VSI}^T \ \Delta \mathbf{X}_{RLC}^T]^T$;

$$\Delta \mathbf{u}_{PMSG-III} = [\Delta T_w \ \Delta P_{G_ref} \ \Delta u_{DC_ref} \ \Delta Q_{g_ref} \ \Delta U_b]^T;$$

$$\mathbf{A}_{PMSG-III} = \begin{bmatrix} \mathbf{A}_M & \mathbf{A}_{G,M} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{A}_{M,G} & \mathbf{A}_G & \mathbf{A}_{S,G} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{G,S} & \mathbf{A}_S & \mathbf{A}_{DC,S} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{A}'_{DC} & \mathbf{A}_{VSI,DC} & \mathbf{A}_{RLC,DC} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{A}_{DC,VSI} & \mathbf{A}_{VSI} & \mathbf{A}_{RLC,VSI} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{A}_{DC,RLC} & \mathbf{A}_{VSI,RLC} & \mathbf{A}'_{RLC} \end{bmatrix}.$$

The unknown submatrices in $\mathbf{A}_{PMSG-III}$ are shown in Appendix C.

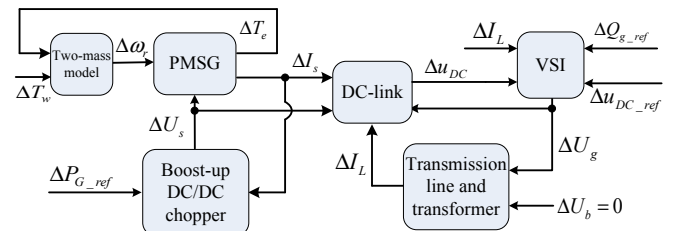


Fig. 7. Block diagram of small-signal model of PMSG-III.

Table 1
Sensitive state variables for SSCI, SSR, SSO or low-frequency oscillation.

Oscillation modes	Classification rules	Sensitive state variables	Frequency band
SSCI	High-speed control of power electronic system and series compensation capacitor	$x1/x2/x3/x4/x5/x6$ and $i/u_C/u_{dc}$	Random
SSR	Series capacitor and the shafting mechanical system	i/u_C and ω/θ	Near to the natural frequency of the shaft
SSO	High-speed electronic switch system and the shafting mechanical system	ω/θ plus some other variables in turbine	Near to the natural frequency of the shaft
low-frequency oscillation	Flux linkage of stator windings	ψ_{ds}/ψ_{qs} dominates, plus some other variables	0.2–2 Hz

Table 2
Eigenvalues of state matrix A_{PMSG-I} .

No.	Eigenvalues	Oscillation frequency (Hz)	Damping ratio
$\lambda_{1,2}$	$-229.81 \pm 448.96i$	71.4540	0.4556
$\lambda_{3,4}$	$-128.39 \pm 240.20i$	38.2284	0.4714
$\lambda_{5,6}$	$-0.3054 \pm 47.54i$	7.5657	0.0064
$\lambda_{7,8}$	$-0.0356 \pm 11.75i$	1.8703	0.0030
λ_9	-7.6412	0	1
λ_{10}	-0.3080	0	1
λ_{11}	-0.0720	0	1
λ_{12}	-1.1601	0	1
λ_{13}	-0.8734	0	1
$\lambda_{14,15}$	$-0.1885 \pm 0.848i$	0.1349	0.2171
λ_{16}	-7.2608×10^{-13}	0	1
λ_{17}	-8.3613×10^{-5}	0	1

Table 3
Participation factors of state variables.

State variables	$\lambda_{1,2}$	$\lambda_{3,4}$	$\lambda_{5,6}$	$\lambda_{7,8}$	$\lambda_{14,15}$
$\Delta\theta_1$	0.0000	0.0000	0.0000	0.0535	0.0000
$\Delta\theta_2$	0.0000	0.0000	0.0000	0.4788	0.0000
$\Delta\omega_1$	0.0000	0.0000	0.0000	0.0535	0.0000
$\Delta\omega_2$	0.0000	0.0000	0.0000	0.4465	0.0000
$\Delta\psi_{ds}$	0.0000	0.0000	0.0000	0.0000	0.5122
$\Delta\psi_{qs}$	0.0000	0.0000	0.0000	0.0323	0.0000
Δi_{Lx}	0.6899	0.0256	0.0527	0.0000	0.0000
Δi_{Ly}	0.7324	0.0786	0.0402	0.0000	0.0000
Δu_{cx}	0.0336	0.5092	0.0224	0.0000	0.0000
Δu_{cy}	0.0640	0.5510	0.0257	0.0000	0.0000
Δu_{DC}	0.0384	0.0270	0.4693	0.0000	0.0000
Δx_1	0.0000	0.0000	0.0000	0.0000	0.5122
Δx_2	0.0000	0.0000	0.0000	0.0000	0.0000
Δx_3	0.0278	0.0236	0.4703	0.0000	0.0000
Δx_4	0.0015	0.0000	0.0012	0.0000	0.0000
Δx_5	0.0000	0.0000	0.0000	0.0000	0.0000
Δx_6	0.0000	0.0000	0.0000	0.0000	0.0000

The participation factors in bold mean that the oscillation modes are highly sensitive to the corresponding state variables. Bold values means strong relation, other values means weak relation.

6. Demonstration examples

Based on the small signal models developed in Section V, 2-MW, 690-V PMSG-based wind turbines are studied in this section. The rated frequency of the power grid is 50 Hz and other parameters are presented in Appendix D. The eigenvalues of three types of PMSGs are calculated from their system state matrix A_{PMSG-I} , $A_{PMSG-II}$ and $A_{PMSG-III}$ respectively in the Matlab.

The process of classifying the different oscillation modes are summarized into 4 steps:

- Step 1: calculating all the oscillation modes based on the unified small signal model for the wind power generation system.
- Step 2: calculating the participation factors between one selected oscillation mode and all state variables.
- Step 3: finding out the sensitive state variables for the selected oscillation mode.

Step 4: classifying the selected oscillation mode into SSCI, SSR, SSO or low-frequency oscillation according to the sensitive state variables, shown in Table 1.

Low-frequency oscillation of PMSG is still open topic now, both in mechanism and physical phenomena. This paper interpret that the low-frequency oscillation of PMSG is dominated by the stator flux linkage and some other variables with oscillation frequency generally between 0.2 and 2 Hz.

6.1. Modal analysis of PMSG-I

6.1.1. Eigenvalues of PMSG-I

The eigenvalues of PMSG-I are obtained from A_{PMSG-I} , shown in Table 2. As seen, there are five oscillation modes and the system is stable after small disturbance because all eigenvalues have negative real parts.

In Table 3, the participation factors which measure the relation between the oscillation modes and state variables are presented. Here, $\Delta i_{Lx,y}$ are the small deviations of i_L in x- and y-axis; $\Delta u_{cx,y}$ are the small deviations of the voltage at series compensation capacitor in x- and y-axis; Δx_i are the intermediate variables of the PI controllers. The participation factors in bold mean that the oscillation modes are highly sensitive to the corresponding state variables. The oscillation modes can be categorized into three types: SSCI, SSO, and low-frequency oscillation.

6.1.1.1. SSCI. The oscillation frequency of $\lambda_{1,2}$ is 71.454 Hz, which is dominated by the x- and y-axis component of the inductor current of the transmission line (participation factor 0.6899 and 0.7324 respectively). The x- and y-axis component of the voltage of the series compensation capacitor also has a certain impact on the mode (participation factor 0.0336 and 0.064 respectively). In addition, the DC-link voltage and immediate variable Δx_3 of the control system of VSI also influence the oscillation mode (participation factor 0.0384 and 0.0278 respectively).

The oscillation frequency of $\lambda_{3,4}$ is 38.2284 Hz and dominated by the x- and y-axis component of the voltage of the series compensation capacitor (participation factor 0.5092 and 0.551 respectively). The x- and y-axis components of the inductor current of the transmission line also have a certain impact on the mode (participation factor 0.0256 and 0.0786 respectively). Similar to $\lambda_{1,2}$, $\lambda_{3,4}$ is also influenced by the DC-link voltage and immediate variable Δx_3 of the control system of VSI (participation factor 0.027 and 0.0236 respectively).

The oscillation frequency of $\lambda_{5,6}$ is 7.5657 Hz and dominated by DC-link voltage and immediate variable Δx_3 of the control system of VSI (participation factor 0.4693 and 0.4703 respectively). The x- and y-axis component of the inductor current of the transmission line and voltage of the series compensation capacitor all influence the mode slightly (participation factor 0.0527, 0.0402, 0.0224 and 0.0257 respectively).

The three oscillation modes $\lambda_{1,2}$, $\lambda_{3,4}$ and $\lambda_{5,6}$ are relevant to

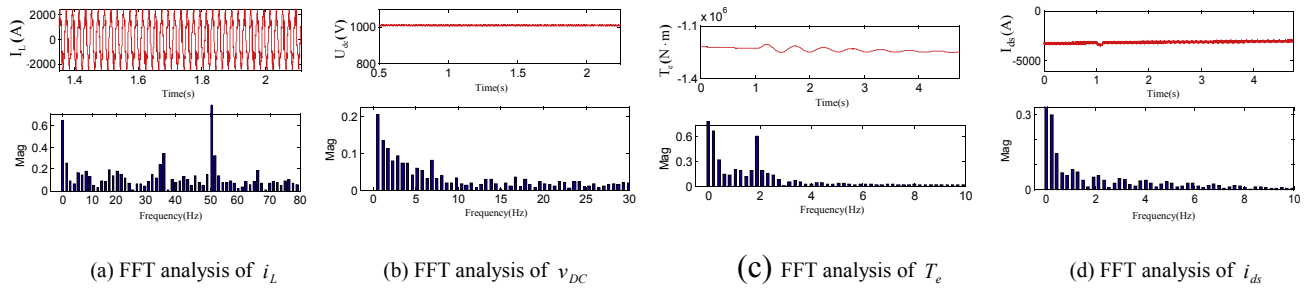


Fig. 8. FFT analysis of oscillation modes of PMSG-I.

Table 4

Prony analysis result of i_L , v_{DC} , T_e and i_{ds} .

Prony object	No.	Oscillation frequency (Hz)	Damping ratio
i_L	$\lambda_{1,2}$	71.6694	0.4551
	$\lambda_{3,4}$	38.2088	0.4774
v_{DC}	$\lambda_{5,6}$	7.5893	0.0065
T_e	$\lambda_{7,8}$	1.8642	0.0030
i_{ds}	$\lambda_{14,15}$	0.1335	0.2151

both the control system of VSI and the electrical system. Thus, they are all SSCI modes.

6.1.1.2. SSO. The oscillation frequency of $\lambda_{7,8}$ is 1.8703 Hz and dominated by the twist angle $\Delta\theta_2$ and angular speed $\Delta\omega_2$ of the generator rotor (participation factor 0.4788 and 0.4465 respectively). Twist angle $\Delta\theta_1$ and angular speed $\Delta\omega_1$ of the wind turbine also influence the oscillation mode (participation factor both 0.0535). The mode is also influenced by the q-axis component of the stator flux linkage $\Delta\psi_{qs}$ (participation factor 0.0323). Thus $\lambda_{7,8}$ is mainly the oscillation mode of the shaft system.

6.1.1.3. Low-frequency oscillation. The oscillation frequency of $\lambda_{14,15}$ is 0.1349 Hz and dominated by the d-axis component $\Delta\psi_{ds}$ of the stator flux linkage and immediate variable Δx_1 of the control system of VSI (participation factor both 0.5122). The participation factors of other state variables in this mode are zero. Thus $\lambda_{14,15}$ is a low-frequency oscillation mode resulted from the interaction between the generator and control system of GSR.

6.1.2. Time-domain simulation

To validate the modal analysis results, the detailed model of PMSG-I is developed in the Matlab/Simulink, shown in Fig. A.1 in Appendix E. The oscillation modes are excited through implementing small disturbances (e.g. voltage and power disturbances). Through FFT and participation factor analysis, the oscillation modes obtained from modal analysis can also be observed from system variables, shown in Fig. 8. Prony analysis results of the time domain simulation are also provided in Table 4.

6.1.2.1. SSCI. From the modal analysis, it is found that $\lambda_{1,2}$ and $\lambda_{3,4}$ are highly relevant to the inductor current and series compensation capacitor voltage of the transmission line. Through conducting FFT analysis on the transmission line current, the results are shown in Fig. 8 (a). As shown in Fig. 8 (a), except for the 50 Hz fundamental component, the harmonic components are mainly in 70 Hz and 38 Hz, which matches 71.454 Hz of $\lambda_{1,2}$ and 38.2284 Hz of $\lambda_{3,4}$ respectively.

From Table 3, $\lambda_{5,6}$ is found to be highly relevant to the voltage of the DC-link voltage and the control system of the VSI. Through conducting FFT analysis on the DC-link voltage, the results are shown in Fig. 8 (b). In Fig. 8 (b), except for the high dc component,

the harmonic components are mainly in 8 Hz, which is consistent with the 7.5657 Hz of $\lambda_{5,6}$.

6.1.2.2. SSO. From Table 3, it is found that $\lambda_{7,8}$ is dominated by the generator rotor. Through conducting FFT analysis on the generator torque, the results are shown in Fig. 8 (c). Except for the dc component, the harmonic components are mainly about 1.9 Hz, which is almost the same with 1.8703 Hz of $\lambda_{7,8}$.

6.1.2.3. Low-frequency oscillation. From Table 3, low-frequency oscillation mode $\lambda_{14,15}$ is highly relevant to $\Delta\psi_{ds}$, which is determined by the d-axis component of the stator current. Through conducting FFT analysis on the d-axis stator current i_{ds} , the results are shown in Fig. 8 (d). The main harmonic is about 0.2 Hz except for the dc component, which is consistent with that of $\lambda_{14,15}$.

6.2. Modal analysis of PMSG-II

6.2.1. Eigenvalues of PMSG-II

The eigenvalues of PMSG-II are obtained from $A_{PMSG-II}$, shown in Table 5. There are six oscillation modes in total and PMSG-II is stable after small disturbance because the real parts of all the eigenvalues are negative.

The participation factors for each oscillation mode are shown in Table 6. The bold participation factors mean the oscillation modes are highly sensitive to the corresponding state variables. The oscillation modes are also categorized into three types: SSCI, SSO, and low-frequency oscillation.

6.2.1.1. SSCI. The oscillation frequency of $\lambda_{1,2}$ is 68.6345 Hz and dominated by the x- and y-axis component of the inductor current of the transmission line (participation factor 0.69 and 0.732 respectively). The mode is also influenced by the x- and y-axis component of the voltage of the series compensation capacitor (participation factor 0.034 and 0.064 respectively). The DC-link voltage and immediate variable Δx_4 of the control system of VSI

Table 5

Eigenvalues of state matrix $A_{PMSG-II}$.

No.	Eigenvalues	Oscillation frequency (Hz)	Damping ratio
$\lambda_{1,2}$	$-220.14 \pm 431.24i$	68.6345	0.4547
$\lambda_{3,4}$	$-140.25 \pm 240.85i$	38.3320	0.5032
$\lambda_{5,6}$	$-0.1239 \pm 47.533i$	7.5652	0.0026
$\lambda_{7,8}$	$-0.026 \pm 11.7514i$	1.8705	0.0024
λ_9	-7.6415	0	1
λ_{10}	-1.5740	0	1
$\lambda_{11,12}$	$-0.0113 \pm 2.2971i$	0.3656	0.0049
λ_{13}	-0.1696	0	1
λ_{14}	-0.0748	0	1
$\lambda_{15,16}$	$-0.762 \pm 0.2129i$	0.0339	0.9631
λ_{17}	-3.5261×10^{-14}	0	1
λ_{18}	-0.0179	0	1

Table 6
Participation factors of state variables.

State variables	$\lambda_{1,2}$	$\lambda_{3,4}$	$\lambda_{5,6}$	$\lambda_{7,8}$	$\lambda_{11,12}$	$\lambda_{15,16}$
$\Delta\theta_1$	0.0000	0.0000	0.0000	0.0530	0.0000	0.9280
$\Delta\theta_2$	0.0000	0.0000	0.0000	0.4790	0.0000	0.9630
$\Delta\omega_1$	0.0000	0.0000	0.0000	0.0530	0.0000	0.9430
$\Delta\omega_2$	0.0000	0.0000	0.0000	0.4460	0.0000	0.1060
$\Delta\psi_{ds}$	0.0000	0.0000	0.0000	0.0000	0.5000	0.0000
$\Delta\psi_{qs}$	0.0000	0.0000	0.0000	0.0320	0.0000	0.6670
Δi_{Lx}	0.6900	0.0260	0.0530	0.0000	0.0000	0.0000
Δi_{Ly}	0.7320	0.0790	0.0400	0.0000	0.0000	0.0000
Δu_{cx}	0.0340	0.5090	0.0220	0.0000	0.0000	0.0000
Δu_{cy}	0.0640	0.5510	0.0260	0.0000	0.0000	0.0000
Δu_{DC}	0.0380	0.0270	0.4690	0.0000	0.0000	0.0000
Δx_1	0.0000	0.0000	0.0000	0.0000	0.0000	1.1740
Δx_2	0.0000	0.0000	0.0000	0.0000	0.0000	0.6500
Δx_3	0.0000	0.0000	0.0000	0.0000	0.5000	0.0000
Δx_4	0.0280	0.0240	0.4700	0.0000	0.0000	0.0000
Δx_5	0.0010	0.0000	0.0010	0.0000	0.0000	0.0000
Δx_6	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Δx_7	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

The participation factors in bold mean that the oscillation modes are highly sensitive to the corresponding state variables. Bold values means strong relation, other values means weak relation.

are also relevant to the oscillation mode (participation factor 0.038 and 0.028 respectively).

The oscillation frequency of $\lambda_{3,4}$ is 38.332 Hz, which is dominated by the x- and y-axis component of the voltage of the series compensation capacitor (participation factor 0.509 and 0.551 respectively). The x- and y-axis component of the inductor current of the transmission line, the DC-link voltage and immediate variable Δx_4 of the control system of VSI are all relevant to the oscillation mode (participation factor 0.026, 0.079, 0.027 and 0.024 respectively).

The oscillation frequency of $\lambda_{5,6}$ is 7.5652 Hz and dominated by DC-link voltage and immediate variable Δx_4 of the control system of VSI (participation factor 0.469 and 0.47 respectively). The mode is also relevant to the x- and y-axis component of the inductor current of the transmission line and voltage of the series compensation capacitor (participation factor 0.053, 0.04, 0.022 and 0.026 respectively).

The three oscillation modes $\lambda_{1,2}$, $\lambda_{3,4}$ and $\lambda_{5,6}$ are all SSCI modes, because they are relevant to both the control system of VSI and the electrical system. Compared to those of PMSG-I, they are quite similar to each other. This is because the topology of PMSG-I is almost the same with PMSG-II, except for the MPPT control in GSR of PMSG-II.

6.2.1.2. SSO. The oscillation frequency of $\lambda_{7,8}$ is 1.8705 Hz, which is dominated by the twist angle $\Delta\theta_2$ and angular speed $\Delta\omega_2$ of the generator rotor (participation factor 0.479 and 0.446 respectively). Twist angle $\Delta\theta_1$ and angular speed $\Delta\omega_1$ of the wind turbine, and the q-axis component of the stator flux linkage $\Delta\psi_{qs}$ are all relevant to

Table 7
Prony analysis result of i_L , v_{DC} , T_e and i_{ds} .

Prony object	No.	Oscillation frequency (Hz)	Damping ratio
i_L	$\lambda_{1,2}$	68.9880	0.4580
	$\lambda_{3,4}$	38.4256	0.5081
v_{DC}	$\lambda_{5,6}$	7.5114	0.0026
T_e	$\lambda_{7,8}$	1.8653	0.0024
i_{ds}	$\lambda_{11,12}$	0.3657	0.0049

the mode. The mode is almost the same with that of PMSG-I, and the damping ratio decreases from 0.003 to 0.0024. Thus, the MPPT control reduces the damping of the shaft system, which influences the stability of the system after small disturbance.

6.2.1.3. Low-frequency oscillation. The oscillation frequency of $\lambda_{11,12}$ is 0.3656 Hz and dominated by the d-axis component $\Delta\psi_{ds}$ of the stator flux linkage and immediate variable Δx_3 of the control system of VSI (participation factor both 0.5). Similar to that of PMSG-I, the participation factors of other state variables in this mode are zero. $\lambda_{11,12}$ is also a low-frequency oscillation mode resulted from the interaction between the generator and control system of GSR. Compared with that of PMSG-I, the oscillation frequency increases while damping ratio decreases apparently. The phenomenon indicates $\lambda_{11,12}$ is influenced a lot by the MPPT control, which may lose stability after small disturbance.

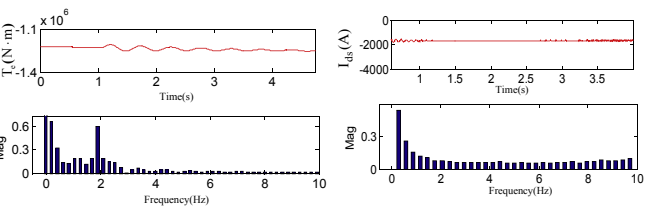
The oscillation frequency of $\lambda_{15,16}$ is 0.0339 Hz, which is dominated by the twist angle $\Delta\theta_1$ and angular speed $\Delta\omega_1$ of the wind turbine (participation factor 0.928 and 0.963), twist angle $\Delta\theta_2$ and angular speed $\Delta\omega_2$ of the generator rotor (participation factor 0.943 and 0.106) and q-axis component of the stator flux linkage $\Delta\psi_{qs}$ (participation factor 0.667). The oscillation mode is also dominated by the immediate variables Δx_1 and Δx_2 of the control system of the GSR. Thus $\lambda_{15,16}$ is a low-frequency oscillation mode resulted from the interaction between the shaft system and the control system after the power signal feedback control is implemented. The oscillation frequency is quite low and damping ratio rather high (0.9631), so the mode is stable after small disturbance.

6.2.2. Time-domain simulation

To validate the modal analysis results, the detailed model of PMSG-II is developed in the Matlab/Simulink, whose topology is the same with Fig. A.1 in Appendix E. Because the oscillation frequency of $\lambda_{15,16}$ is so low and the mode is quite stable, the influence of the mode on the system stability is not studied further. FFT and Prony analysis results are shown in Fig. 9 and Table 7.

6.2.2.1. SSCI. From Table 6, $\lambda_{1,2}$ and $\lambda_{3,4}$ are found highly relevant to the inductor current and series compensation capacitor voltage of the transmission line. Through conducting FFT analysis on the transmission line current, the results are shown in Fig. 9 (a). In Fig. 9 (a), except for the fundamental 50 Hz component, the harmonic

the mode. The mode is almost the same with that of PMSG-I, and the damping ratio decreases from 0.003 to 0.0024. Thus, the MPPT control reduces the damping of the shaft system, which influences the stability of the system after small disturbance.



(a) FFT analysis of i_L

(b) FFT analysis of v_{DC}

(c) FFT analysis of T_e

(d) FFT analysis of i_{ds}

Fig. 9. FFT analysis of oscillation modes of PMSG-II.

components are mainly in 67 Hz and 38 Hz, which is consistent with 68.6345 Hz of $\lambda_{1,2}$ and 38.332 Hz of $\lambda_{3,4}$.

From modal analysis, $\lambda_{5,6}$ is found to be highly relevant to the DC-link voltage. Through conducting FFT analysis on the DC-link voltage, the results are shown in Fig. 9 (b). Except for the dc component, there are also harmonic components in 8 Hz, which is almost the same with the oscillation frequency of $\lambda_{5,6}$.

6.2.2.2. SSO. Similar to PMSG-I, $\lambda_{7,8}$ is also dominated by the generator rotor. Through conducting FFT analysis on the generator torque, the results are shown in Fig. 9 (c). From Fig. 9 (c), it is found that except for the dc component, the harmonic components are mainly about 1.9 Hz, which is almost the same with 1.8705 Hz of $\lambda_{7,8}$.

6.2.2.3. Low-frequency oscillation. Similar to PMSG-I, the low-frequency oscillation mode $\lambda_{11,12}$ is also analyzed through d-axis stator current Δi_{ds} . After conducting FFT analysis on Δi_{ds} , the results are shown in Fig. 9 (d). As shown, the harmonic component is mainly 0.5 Hz except for the dc component, which is consistent with 0.3656 Hz of $\lambda_{11,12}$.

6.3. Modal analysis of PMSG-III

6.3.1. Eigenvalues of PMSG-III

The eigenvalues of PMSG-III are calculated from $A_{PMSG-III}$, shown in Table 8. As seen, there are six oscillation modes and the system is stable after small disturbance for all eigenvalues have negative real parts. The oscillation frequency of $\lambda_{5,6}$ is 50 Hz, which reflects the rated system frequency. The oscillation frequency of $\lambda_{12,13}$ is low and damping ratio rather high, so the mode is rather stable after small disturbances. The two modes are neglected during the research.

The participation factors for each oscillation mode are shown in Table 9. Participation factors in bold mean the oscillation modes are highly sensitive to the corresponding state variables. Unlike the other two types of PMSGs, the oscillation modes of PMSG-III are categorized into two types: SSCI and SSO.

6.3.1.1. SSCI. The oscillation frequency of $\lambda_{1,2}$ is 59.9272 Hz and dominated by the x- and y-axis component of the inductor current of the transmission line (participation factor 1.0433 and 1.573 respectively). The mode is also dominated by the x- and y-axis component of the voltage of the series compensation capacitor, and the DC-link voltage (participation factor 0.1394, 0.3094 and 0.3345 respectively). The immediate variable Δx_3 of the control system of VSI is also relevant to the oscillation mode (participation factor 0.0041).

The oscillation frequency of $\lambda_{3,4}$ is 40.8733 Hz and dominated by

Table 8
Eigen values of state matrix $A_{PMSG-III}$.

No.	Eigen values	Oscillation frequency (Hz)	Damping ratio
$\lambda_{1,2}$	$-474.5 \pm 376.53i$	59.9272	0.7833
$\lambda_{3,4}$	$-127.17 \pm 256.81i$	40.8733	0.4437
$\lambda_{5,6}$	$-0.0748 \pm 314.15i$	50	2.38×10^{-4}
$\lambda_{7,8}$	$-9.1947 \pm 7.4603i$	1.1878	0.7764
$\lambda_{9,10}$	$-0.0266 \pm 12.127i$	1.9302	0.0022
λ_{11}	-0.1667	0	1
$\lambda_{12,13}$	$-0.099 \pm 0.0019i$	2.998×10^{-4}	0.9998
λ_{14}	-0.0176	0	1
λ_{15}	-0.0067	0	1
λ_{16}	-3.3726×10^{-4}	0	1
λ_{17}	-2.4196×10^{-13}	0	1

Table 9
Participation factors of state variables.

State variables	$\lambda_{1,2}$	$\lambda_{3,4}$	$\lambda_{5,6}$	$\lambda_{7,8}$	$\lambda_{9,10}$
$\Delta \theta_1$	0.0000	0.0000	0.0000	0.0000	0.0472
$\Delta \theta_2$	0.0000	0.0000	0.0000	0.0000	0.4528
$\Delta \omega_1$	0.0000	0.0000	0.0000	0.0000	0.0472
$\Delta \omega_2$	0.0000	0.0000	0.0000	0.0000	0.4529
$\Delta \psi_{ds}$	0.0000	0.0000	0.5000	0.0000	0.0000
$\Delta \psi_{qs}$	0.0000	0.0000	0.5000	0.0000	0.0000
Δi_{Lx}	1.0433	0.1934	0.0000	0.0067	0.0000
Δi_{Ly}	1.5730	0.6366	0.0000	0.0320	0.0000
Δu_{cx}	0.1394	0.6286	0.0000	0.0123	0.0000
Δu_{cy}	0.3094	0.8384	0.0000	0.0022	0.0000
Δu_{DC}	0.3345	0.1872	0.0000	0.8518	0.0000
Δx_1	0.0000	0.0000	0.0000	0.0000	0.0000
Δx_2	0.0000	0.0000	0.0000	0.0000	0.0000
Δx_3	0.0041	0.0049	0.0000	0.8222	0.0000
Δx_4	0.0000	0.0000	0.0000	0.0018	0.0000
Δx_5	0.0000	0.0000	0.0000	0.0000	0.0000
Δx_6	0.0000	0.0000	0.0000	0.0000	0.0000

The participation factors in bold mean that the oscillation modes are highly sensitive to the corresponding state variables. Bold values means strong relation, other values means weak relation.

the x- and y-axis component of the inductor current of the transmission line, x- and y-axis component of the voltage of the series compensation capacitor and the DC-link voltage (participation factor 0.1934, 0.6366, 0.6286, 0.8384 and 0.1872 respectively). The mode is also relevant to the immediate variable Δx_3 of the control system of VSI (participation factor 0.0049).

The oscillation frequency of $\lambda_{7,8}$ is 1.1878 Hz, which is dominated by the DC-link voltage and immediate variable Δx_3 of the control system of VSI (participation factor 0.8518 and 0.8222 respectively). The mode is also influenced by the y-axis component of the inductor current of the transmission line (participation factor 0.032) and x-axis component of the voltage of the series compensation capacitor (participation factor 0.0123).

The three modes $\lambda_{1,2}$, $\lambda_{3,4}$ and $\lambda_{7,8}$ are relevant to both the control system of VSI and the electrical system. Thus they are all SSCI modes. The SSCI modes of PMSG-III are quite different from those of the other two types of PMSGs after the topology is changed.

6.3.1.2. SSO. The oscillation frequency of $\lambda_{9,10}$ is 1.9302 Hz, which is dominated by the twist angle $\Delta \theta_2$ and angular speed $\Delta \omega_2$ of the generator rotor (participation factor 0.4528 and 0.4529 respectively). Twist angle $\Delta \theta_1$ and angular speed $\Delta \omega_1$ of the wind turbine are all relevant to the mode (participation factor both 0.0472). Compared with the SSO modes of the other two types of PMSGs, the oscillation frequency is higher, which results from the impact of different model connection in different topologies.

6.3.2. Time-domain simulation

To validate the modal analysis results, the detailed model of PMSG-III is developed in the Matlab/Simulink, shown in Fig. A.2 in Appendix E. Through FFT and participation factor analysis, the oscillation modes obtained from modal analysis can be also observed from corresponding variables, shown in Fig. 10. Prony analysis results are also provided in Table 10.

6.3.2.1. SSCI. From Table 9, $\lambda_{1,2}$ and $\lambda_{3,4}$ are highly relevant to the inductor current and series compensation capacitor voltage of the transmission line. Through conducting FFT analysis on the transmission line current, the results are shown in Fig. 10 (a). Except for the 50 Hz fundamental component, the harmonic components are mainly 40 Hz and 60 Hz, which complies with the results of $\lambda_{1,2}$ and $\lambda_{3,4}$.

From modal analysis, $\lambda_{7,8}$ is highly relevant with the DC-link

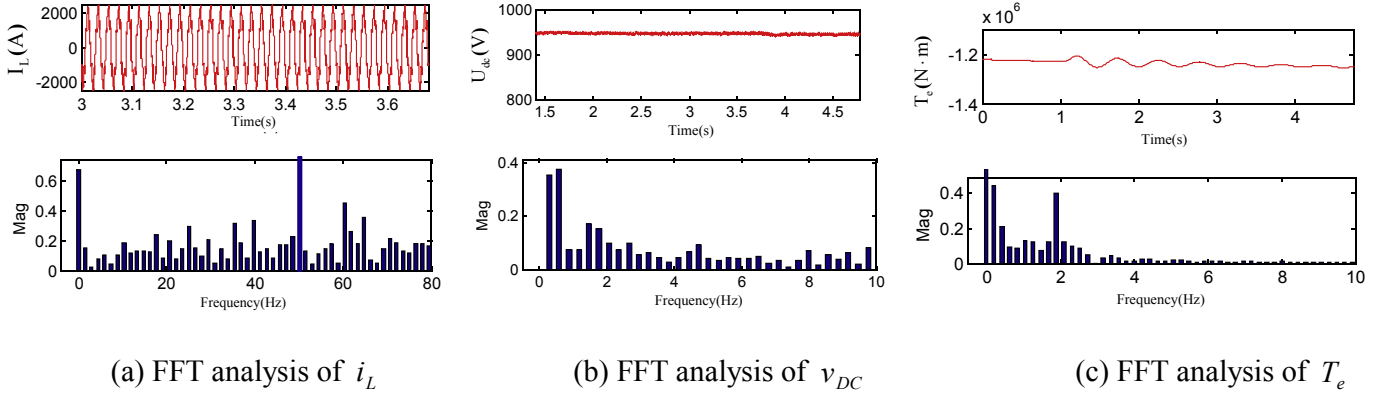


Fig. 10. FFT analysis of oscillation modes of PMSG-III.

Table 10

Prony analysis result of i_L , v_{DC} , T_e and i_{ds} .

Prony object	No.	Oscillation frequency (Hz)	Damping ratio
i_L	$\lambda_{1,2}$	59.6186	0.7801
	$\lambda_{3,4}$	40.8438	0.4494
v_{DC}	$\lambda_{7,8}$	1.1833	0.7755
T_e	$\lambda_{9,10}$	1.9339	0.0022

voltage. Through conducting FFT analysis on the DC-link voltage, the results are shown in Fig. 10 (b). Except for the dc component, the harmonic components in Fig. 10 (b) are mainly between 1 and 2 Hz, which is consistent with the oscillation frequency of $\lambda_{7,8}$.

6.3.2.2. SSO. Similar to the other two types of PMSGs, $\lambda_{9,10}$ is also dominated by the generator rotor. Through conducting FFT analysis on the generator torque, the results are shown in Fig. 10 (c). From Fig. 10 (c), it is found that except for the dc component, the harmonic components are mainly about 1.95 Hz, which is almost the same with 1.9302 Hz of $\lambda_{9,10}$.

6.4. Influence of different initial values

Initial operation values and control system parameters are two important parameters affecting oscillation modes and their influence naturally need to be analyzed. However, space limitation should be considered because there are 12 control parameters for PMSG-I, 14 control parameters for PMSG-II and 12 control parameters for PMSG-III. A full analysis of control system parameters will take the very great length. In addition, initial operation values have a greater influence on the oscillation modes compared with control system parameters.

So, at first, the control parameters of the model are determined from wind-turbine makers. Then through linearising the small signal models of PMSGs at different initial operating points, the influence of different initial values on the oscillation modes can be studied correspondingly. During the research, the output power is varied from 0.05 p.u. to 1 p.u. to find the general law.

6.4.1. Influence of different initial values on PMSG-I

When the output power is varied from 0.05 p.u. to 1 p.u., the low-frequency oscillation mode is almost irrelevant with the initial values and thus only the variation of SSCI and SSO modes of PMSG-I is presented in Fig. 11.

In Fig. 11 (a), both the oscillation frequency and damping ratio of $\lambda_{1,2}$ increase with the output power increasing. When the output power is 0.8 p.u. and 0.9 p.u., the oscillation frequency and damping ratio of $\lambda_{1,2}$ change suddenly. For $\lambda_{3,4}$ in Fig. 11 (b), the oscillation frequency of $\lambda_{3,4}$ increases while the damping ratio of $\lambda_{3,4}$ decreases with the output power increasing. Similar to $\lambda_{1,2}$, the oscillation frequency and damping ratio of $\lambda_{3,4}$ also change suddenly when the output power is 0.8 p.u. and 0.9 p.u. In Fig. 11 (c), when the output power is 0.05–0.25 p.u., the damping ratio of $\lambda_{5,6}$ is negative, which means the mode lose the stability. The oscillation frequency and damping ratio of $\lambda_{5,6}$ also change suddenly when the output power is 0.8 p.u. and 0.9 p.u. When the output power is 0.9 p.u., the oscillation mode even becomes evanescent mode. At other output power values, both the oscillation frequency and damping ratio increases gradually with the increasing power. When the output power increases, the oscillation frequency decreases while the damping ratio increases for the SSO mode $\lambda_{7,8}$, as shown in Fig. 11 (d).

6.4.2. Influence of different initial values on PMSG-II

When the output power is varied from 0.05 p.u. to 1 p.u., the low-frequency oscillation mode $\lambda_{11,12}$ is almost irrelevant with the initial values and thus only the variation of SSCI, SSO and low-frequency oscillation mode $\lambda_{15,16}$ of PMSG-II is presented in Fig. 12.

As shown in Fig. 12, the variation trend of SSCI modes (i.e. $\lambda_{1,2}$, $\lambda_{3,4}$ and $\lambda_{5,6}$) of PMSG-II with initial values is similar to that of PMSG-I shown in Fig. 11. Then the explanation of Fig. 12 (a)–(c) is omitted here. In Fig. 12 (d), the damping ratio of the SSO mode increases and the oscillation frequency decreases with the output power increasing. When PMSG-II is operated at the nominal state, the damping ratio of the SSO mode reaches the maximum. As for the low-frequency mode $\lambda_{15,16}$ in Fig. 12 (e), the damping ratio

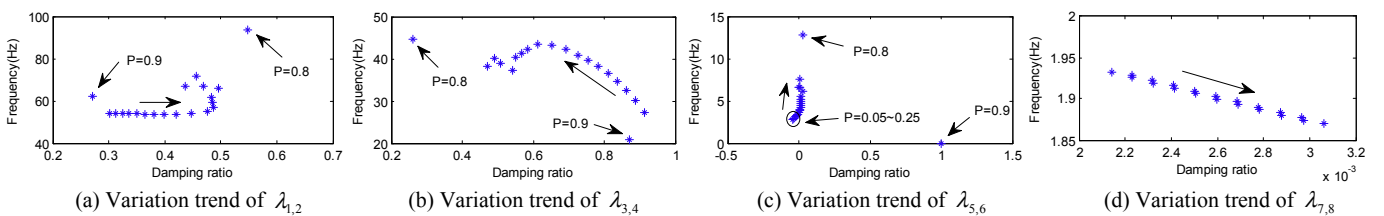


Fig. 11. Variation trend of SSCI and SSO modes with initial values of PMSG-I.

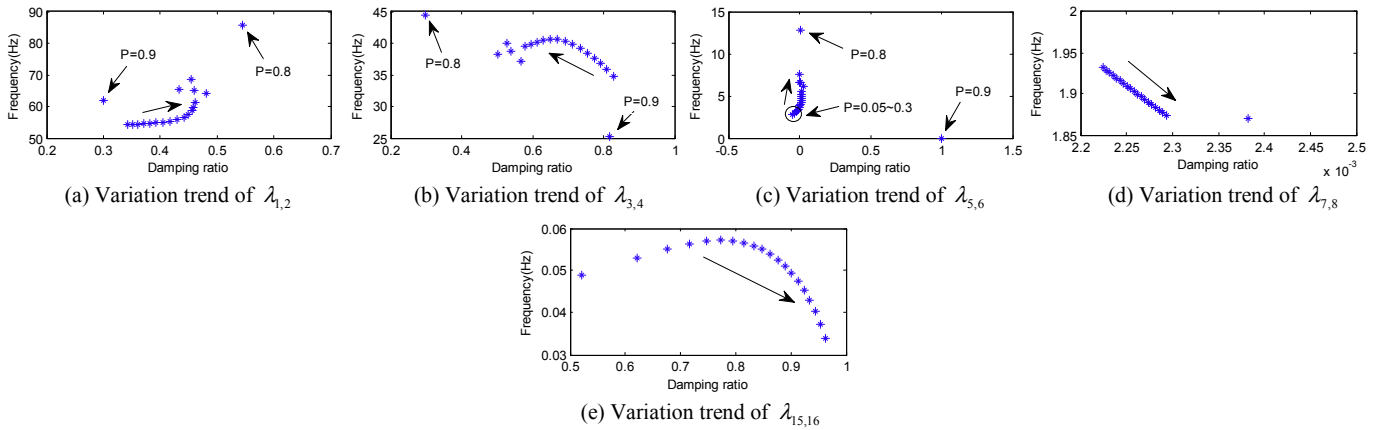


Fig. 12. Variation trend of oscillation modes with initial values of PMSG-II.

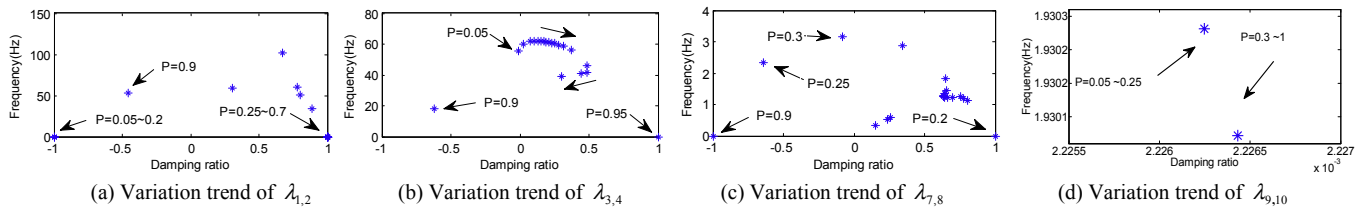


Fig. 13. Variation trend of SSCI modes with initial values of PMSG-III.

increases with the output power increasing whereas the oscillation frequency increases first and then decreases. Thus, the mode is more stable when the output power is high.

6.4.3. Influence of different initial values on PMSG-III

When the output power is varied from 0.05 p.u. to 1 p.u., the SSO mode is almost irrelevant with the initial values and thus only the variation trend of SSCI mode of PMSG-III is presented in Fig. 13.

As shown in Fig. 13, the initial values will influence both the oscillation frequencies and damping ratios of SSCI modes and even make them unstable. Compared to the other two types of PMSGs, the regular variation trend of the oscillation modes with the output power is also lost. In Fig. 13 (a), when the output power ranges from 0.05 to 0.2 p.u., $\lambda_{1,2}$ is in evanescent mode, where the damping ratio is negative and the mode is unstable. $\lambda_{1,2}$ is in the stable evanescent mode when the output power is between 0.25 p.u. and 0.7 p.u. When the output power is at 0.9 p.u., $\lambda_{1,2}$ is in the unstable oscillation mode. When the output power is 0.05 p.u. and 0.9 p.u., the damping ratio of $\lambda_{3,4}$ is negative and the mode is unstable correspondingly, shown in Fig. 13 (b). When the output power is 0.95 p.u., $\lambda_{3,4}$ turns to the stable evanescent mode. Then, $\lambda_{3,4}$ is in the stable oscillation modes, when the output power is at other values. In Fig. 13 (c), when the output power is 0.2 p.u. and 0.9 p.u., $\lambda_{7,8}$ is in the stable and unstable evanescent mode respectively. While $\lambda_{7,8}$ is in the unstable oscillation modes when the output power is 0.25 p.u. and 0.3 p.u. When the output power is at other values, $\lambda_{7,8}$ is in the stable oscillation modes.

7. Conclusions

In this paper, unified small-signal models are established for PMSG-based WTGS with different converter topologies. Based on the small-signal models through eigenvalue and modal analysis, the relation between different oscillation modes and system variables is extensively studied. From the demonstration, the

conclusions are summarized as follows.

- For PMSG-based WTGS with back-to-back PWM converter, there are mainly SSCI, SSO and low-frequency oscillation modes even different control schemes are adopted for GSR. Whereas, there are only SSCI and SSO oscillation modes for PMSGs with DC boost converter systems. PMSG-III is found to be superior to PMSG-I/II in fewer oscillation modes.
- SSCI modes are rather severe in the three types of PMSGs. As for PMSG-I and PMSG-II, there are three oscillation modes in their SSCI mode, where the damping ratio of the 7.5 Hz mode is rather low and may be easily excited after small disturbances. There are also three oscillation modes in the SSCI mode of PMSG-III, and the damping ratio is relatively high. Thus, PMSG-III is also correspondingly more stable than other two types of PMSGs after small disturbances.
- In the three types of PMSGs, the initial values influence both oscillation frequencies and damping ratios. With variations of output power, a sudden change may happen to the oscillation modes at certain power points and even render the oscillation modes into evanescent modes. At the same time, the PMSGs may lose small signal stability when the damping ratio is negative at a certain power point.
- The proposed model is essentially derived from Lyapunov First Theory. It needs to model the whole system with perfect precision. All model parameters must be recognized from wind-turbine makers. Compared to Prony method, it takes longer calculating time. In addition, eigenvalue calculation of the system state matrix can be very complex when applied to wind farms, i.e., ‘curse of dimensionality’.

Acknowledgement

This work was supported by the National Natural Science Foundation of China (51277119 and 51677114).

Appendix A. Unknown submatrices in $\mathbf{A}_{PMSC-I}$

$$\begin{aligned}
\mathbf{A}_{G_M} &= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3n_p\psi_{PM}}{2J_1L_{qs}} \end{bmatrix}^T; \quad \mathbf{A}_{M_G} = \begin{bmatrix} 0 & 0 & 0 & \psi_{qs0} + L_{qs}i_{ds0} \\ 0 & 0 & 0 & \psi_{PM} + \psi_{ds0} \end{bmatrix}; \quad \mathbf{A}'_G = \begin{bmatrix} \frac{K_{p2}}{L_{ds}} & 0 \\ 0 & 0 \end{bmatrix}; \quad \mathbf{A}_{GSR_G} = \begin{bmatrix} 0 & -K_{i2} \\ -K_{i1} & 0 \end{bmatrix}; \quad \mathbf{A}_{G_GSR} = \begin{bmatrix} 0 & 0 \\ \frac{1}{L_{ds}} & 0 \end{bmatrix}; \\
\mathbf{A}_{DC_GSR} &= \mathbf{0}_{2 \times 1}; \quad \mathbf{A}_{M_DC} = \begin{bmatrix} 0 & 0 & 0 & -L_{qs}i_{ds0}^2 - \psi_{PM}i_{qs0} \end{bmatrix}; \quad \mathbf{A}_{G_DC} = \begin{bmatrix} \frac{\omega_b L_{ds} i_{qs0} - (K_{p2} - R_s) i_{ds0} - u_{ds0}}{L_{ds}} & \frac{R_s i_{qs0} - \omega_b L_{qs} i_{ds0} - u_{qs0}}{L_{qs}} \end{bmatrix}; \\
\mathbf{A}_{GSR_DC} &= \begin{bmatrix} K_{i1} i_{qs0} & K_{i2} i_{ds0} \end{bmatrix}; \quad \mathbf{A}'_{DC} = \begin{bmatrix} -K_{p3} K_{p4} i_{gd0} \end{bmatrix}; \quad \mathbf{A}_{RLC_VSI} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ -u_{qg0} & u_{dg0} & 0 & 0 \\ -K_{p5} u_{qg0} & K_{p5} u_{dg0} - 1 & 0 & 0 \end{bmatrix}; \quad \mathbf{A}_{DC_RLC} = \begin{bmatrix} -\frac{K_{p3} K_{p4} \omega_b}{X_{TL}} & 0 & 0 & 0 \end{bmatrix}^T; \\
\mathbf{A}_{VSI_DC} &= \begin{bmatrix} K_{p4} K_{i3} i_{dg0} \left(\frac{K_{p5} K_{p6} i_{qg0}^2}{1 + K_{p5} K_{p6} u_{qg0}} + 1 \right) & K_{i4} i_{dg0} \left(\frac{K_{p5} K_{p6} i_{qg0}^2}{1 + K_{p5} K_{p6} u_{qg0}} + 1 \right) & \frac{K_{p5} K_{i5} i_{qg0}}{1 + K_{p5} K_{p6} u_{qg0}} & \frac{K_{i6} i_{qg0}}{1 + K_{p5} K_{p6} u_{qg0}} \end{bmatrix}; \quad \mathbf{A}_{RLC_DC} = \begin{bmatrix} -K_{p3} i_{gd0} \\ \frac{K_{p5} K_{p6} u_{qg0} i_{qg0}}{1 + K_{p5} K_{p6} u_{qg0}} + u_{dg0} \frac{(K_{p5} u_{dg0} - 1) K_{p6} i_{qg0}}{1 + K_{p5} K_{p6} u_{qg0}} + u_{qg0} 0 \end{bmatrix}; \\
\mathbf{A}_{VSI_RLC} &= \begin{bmatrix} \frac{\omega_b K_{i3} K_{p4}}{X_{TL}} & \frac{\omega_b K_{i4}}{X_{TL}} & 0 & 0 \\ \frac{\omega_b K_{p5} K_{p6} i_{qg0} K_{i3} K_{p4}}{(1 + K_{p5} K_{p6} u_{qg0}) X_{TL}} & \frac{\omega_b K_{p5} K_{p6} i_{qg0} K_{i4}}{(1 + K_{p5} K_{p6} u_{qg0}) X_{TL}} & \frac{\omega_b K_{p5} K_{i5}}{(1 + K_{p5} K_{p6} u_{qg0}) X_{TL}} & \frac{\omega_b K_{i6}}{(1 + K_{p5} K_{p6} u_{qg0}) X_{TL}} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}; \\
\mathbf{A}'_{RLC} &= \begin{bmatrix} \frac{\omega_b}{X_{TL}} (-R_{TL} - K_{p5} K_{p6} u_{qg0}) & \omega_b (1 + \frac{(K_{p5} u_{dg0} - 1) K_{p6}}{X_{TL}}) & -\frac{\omega_b}{X_{TL}} & 0 \\ -\omega_b \left(1 + \frac{K_{p5} K_{p6} u_{qg0}}{X_{TL} (1 + K_{p5} K_{p6} u_{qg0})} \right) & \frac{\omega_b}{X_{TL}} \left(\frac{(K_{p5} u_{dg0} - 1) K_{p6}}{1 + K_{p5} K_{p6} u_{qg0}} - R_{TL} \right) & 0 & -\frac{\omega_b}{X_{TL}} \\ -\omega_b X_C & 0 & 0 & \omega_b \\ 0 & -\omega_b X_C & -\omega_b & 0 \end{bmatrix}.
\end{aligned}$$

Appendix B. Unknown submatrices in $\mathbf{A}_{PMSC-II}$

$$\begin{aligned}
\mathbf{A}_{M_G} &= \begin{bmatrix} 0 & 0 & 0 & L_{qs} i_{qs0} + \psi_{qs0} \\ 0 & 0 & 0 & \frac{K_{p1} K_{p2} i_{ds0} i_{qs0} L_{qs} + \psi_{PM} - L_{ds} i_{ds0}}{1 - K_{p1} K_{p2} i_{qs0}} \end{bmatrix}; \quad \mathbf{A}'_{GSR} = \begin{bmatrix} \frac{K_{p1} K_{im} i_{qs0}}{1 - K_{pm} K_{p1} i_{qs0}} & \frac{K_{i1} i_{qs0}}{1 - K_{pm} K_{p1} i_{qs0}} & \frac{K_{i2} i_{ds0}}{1 - K_{pm} K_{p1} i_{qs0}} \\ \frac{K_{pm} K_{p1} K_{im} i_{qs0}}{1 - K_{pm} K_{p1} i_{qs0}} + K_{i1} & \frac{K_{pm} K_{i1} i_{qs0}}{1 - K_{pm} K_{p1} i_{qs0}} & \frac{K_{pm} K_{i2} i_{ds0}}{1 - K_{pm} K_{p1} i_{qs0}} \\ 0 & 0 & 0 \end{bmatrix}; \\
\mathbf{A}'_G &= \begin{bmatrix} -\frac{K_{p2}}{L_{ds}} & 0 \\ \frac{\omega_b L_{ds} - K_{pm} K_{p1} i_{ds0} (K_{p2} - R_s) - K_{pm} K_{p1} u_{ds0}}{(1 - K_{pm} K_{p1} i_{qs0}) L_{ds}} - \omega_b & -\frac{K_{pm} K_{p1} i_{ds0} \omega_b L_{qs} + K_{pm} K_{p1} u_{qs0} + K_{p1} - R_s}{(1 - K_{pm} K_{p1} i_{qs0}) L_{qs}} - \frac{R_s}{L_{qs}} \end{bmatrix}; \\
\mathbf{A}_{GSR_G} &= \begin{bmatrix} 0 & 0 & -K_{i2} \\ -K_{p1} K_{im} & -K_{i1} & -K_{pm} K_{p1} i_{ds0} K_{i2} \end{bmatrix}; \quad \mathbf{A}_{GSR_DC} = \begin{bmatrix} \frac{K_{p1} K_{im} i_{qs0}}{1 - K_{pm} K_{p1} i_{qs0}} & \frac{K_{i1} i_{qs0}}{1 - K_{pm} K_{p1} i_{qs0}} & \frac{K_{i2} i_{ds0}}{1 - K_{pm} K_{p1} i_{qs0}} \end{bmatrix}; \\
\mathbf{A}_{G_GSR} &= \begin{bmatrix} 0 & 0 & 0 & -L_{qs} i_{ds0} i_{qs0} - \frac{(K_{pm} K_{p1} i_{ds0} i_{qs0} L_{qs} + \psi_{PM} - L_{sd} i_{ds0}) i_{qs0}}{1 - K_{pm} K_{p1} i_{qs0}} \\ 0 & 0 & 0 & K_{pm} \left[-L_{qs} i_{ds0} i_{qs0} - \frac{(K_{pm} K_{p1} i_{ds0} i_{qs0} L_{qs} + \psi_{PM} - L_{sd} i_{ds0}) i_{qs0}}{1 - K_{pm} K_{p1} i_{qs0}} \right] \\ 0 & 0 & 0 & 0 \end{bmatrix}; \\
\mathbf{A}_{M_DC} &= \begin{bmatrix} 0 & 0 & 0 & -L_{qs} i_{ds0} i_{qs0} - \frac{(K_{pm} K_{p1} i_{ds0} i_{qs0} L_{qs} + \psi_{PM} - L_{sd} i_{ds0}) i_{qs0}}{1 - K_{pm} K_{p1} i_{qs0}} \end{bmatrix}; \\
\mathbf{A}_{G_GSR} &= \begin{bmatrix} \frac{\omega_b L_{ds} i_{qs0} - (K_{p2} - R_s) i_{ds0} (1 - 2K_{pm} K_{p1} i_{qs0}) - 2K_{pm} K_{p1} u_{ds0} i_{qs0} + u_{ds0}}{(1 - K_{pm} K_{p1} i_{qs0}) L_{ds}} & \frac{\omega_b L_{qs} i_{ds0} + (K_{p1} - R_s) i_{qs0} + u_{qs0}}{(1 - K_{pm} K_{p1} i_{qs0}) L_{qs}} \\ K_{pm} \left[\frac{\omega_b L_{ds} i_{qs0} - (K_{p2} - R_s) i_{ds0} (1 - 2K_{pm} K_{p1} i_{qs0}) - 2K_{pm} K_{p1} u_{ds0} i_{qs0} + u_{ds0}}{(1 - K_{pm} K_{p1} i_{qs0}) L_{ds}} \right] & K_{pm} \left[\frac{\omega_b L_{qs} i_{ds0} + (K_{p1} - R_s) i_{qs0} + u_{qs0}}{(1 - K_{pm} K_{p1} i_{qs0}) L_{qs}} \right] \\ 0 & \frac{1}{L_{qs}} \end{bmatrix}; \\
\mathbf{A}_{DC_GSR} &= \mathbf{0}_{3 \times 1}; \quad \mathbf{A}_{G_DC} = \begin{bmatrix} \frac{\omega_b L_{ds} i_{qs0} - (K_{p2} - R_s) i_{ds0} (1 - 2K_{pm} K_{p1} i_{qs0}) - 2K_{pm} K_{p1} u_{ds0} i_{qs0} + u_{ds0}}{(1 - K_{pm} K_{p1} i_{qs0}) L_{ds}} & \frac{\omega_b L_{qs} i_{ds0} + (K_{p1} - R_s) i_{qs0} + u_{qs0}}{(1 - K_{pm} K_{p1} i_{qs0}) L_{qs}} \end{bmatrix}.
\end{aligned}$$

$A_{G-M}, A'_{DC}, A_{VSI-DC}(A_{DC-VSI}), A_{RLC-DC}(A_{DC-RLC}), A_{RLC-VSI}(A_{VSI-RLC})$ and A_{RLC} are the same with those of A_{PMSG-I} .

Appendix C. Unknown submatrices in $A_{PMSG-III}$

$A_{M-G} = \begin{bmatrix} 0 & 0 & 0 & \psi_{qs0} \\ 0 & 0 & 0 & \psi_{ds0} \end{bmatrix}$; $A_{S-G} = A_{G-S} = \mathbf{0}_{2 \times 2}$; $A_{S-DC} = \mathbf{0}_{1 \times 2}$;
 $A_{DC-S} = \mathbf{0}_{2 \times 1}$; $A_{G-DC} = \begin{bmatrix} \frac{u_{ds0}}{L_{ds}} & \frac{u_{qs0}}{L_{qs}} \end{bmatrix}$; $A_{G-M}, A'_{DC}, A_{VSI-DC}(A_{DC-VSI}),$
 $A_{DC-RLC}(A_{RLC-DC}), A_{VSI-RLC}(A_{RLC-VSI})$ and A'_{RLC} are the same as those
of A_{PMSG-I} .

Appendix D. Parameters of the PMSGs

Wind turbine: $J_1 = 6.2506 \times 10^6 \text{ kg m}^2, J_2 = 6.511 \times 10^5 \text{ kg m}^2,$
 $K_{12} = 8.6727 \times 10^7 \text{ Nm/rad}, D_1 = 1.8 \text{ p.u.}, D_2 = 0.8 \text{ p.u.}, D_{12} = 0.$
PMSG: $n_p = 48, P = 2 \text{ MW}, L_d = L_q = 0.010052 \text{ H}, R_s = 0.0236 \Omega,$
 $\psi_{PM} = 1.05 \text{ p.u.}$

Control parameters for converter system of PMSG-I: $K_{p1} = 5,$
 $K_{i1} = 10; K_{p2} = 2, K_{i2} = 0.03; K_{p3} = 0.05, K_{i3} = 20; K_{p4} = 6.5, K_{i4} = 50;$
 $K_{p5} = 0.04, K_{i5} = 0.01; K_{p6} = 20, K_{i6} = 7.$

Control parameters for converter system of PMSG-II: $K_{pm} = 0.2,$
 $K_{im} = 0.3; K_{p1} = 0.001, K_{i1} = 0.01; K_{p2} = 0.3, K_{i2} = 70; K_{p3} = 0.05,$
 $K_{i3} = 20; K_{p4} = 6.5, K_{i4} = 50; K_{p5} = 0.03, K_{i5} = 0.01; K_{p6} = 17, K_{i6} = 4.$

Control parameters for converter system of PMSG-III:
 $K_{p1} = 0.061, K_{i1} = 0.0002; K_{p2} = 0.6, K_{i2} = 0.04; K_{p3} = 27, K_{i3} = 200;$
 $K_{p4} = 30, K_{i4} = 5; K_{p5} = 100, K_{i5} = 10; K_{p6} = 10, K_{i6} = 1.$

Transmission line: $R_T = 0.007 \text{ p.u.}, X_T = 0.044 \text{ p.u.}, R_L = 0.8402$
 $\text{p.u.}, X_L = 8.402 \text{ p.u.}$

Appendix E. Detailed models of PMSGs

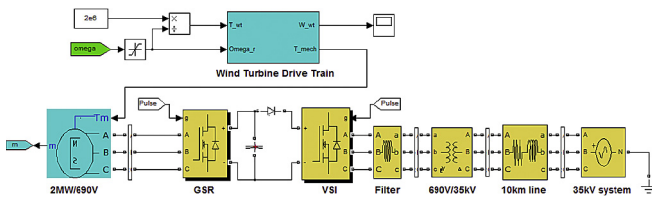


Fig. A.1. Detailed model of PMSG-I/II in Matlab/Simulink.

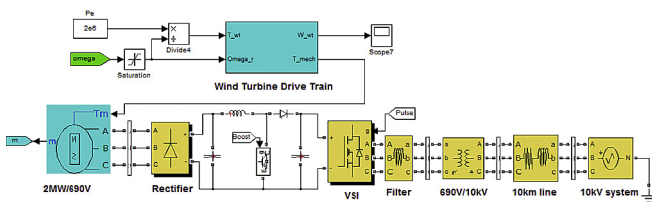


Fig. A.2. Detailed model of PMSG-III in Matlab/Simulink.

References

- [1] Alona Armstrong, et al., Wind farm and solar park effects on plant–soil carbon cycling: uncertain impacts of changes in ground-level microclimate, *Glob. change Biol.* 20 (6) (2014) 1699–1706.
- [2] Jason M. Kennedy, et al., Validation of fixed speed induction generator models for inertial response using wind farm measurements, *Power Syst. IEEE Trans.* 26 (3) (2011) 1454–1461.
- [3] Jesús López, et al., Ride through of wind turbines with doubly fed induction generator under symmetrical voltage dips, *Industrial Electron. IEEE Trans.* 56 (10) (2009) 4246–4254.
- [4] Seifeddine Benelghali, et al., Experimental validation of a marine current turbine simulator: application to a permanent magnet synchronous generator-based system second-order sliding mode control, *Industrial Electron. IEEE Trans.* 58 (1) (2011) 118–126.
- [5] Marco Liserre, et al., Overview of multi-MW wind turbines and wind parks, *IEEE Trans. Industrial Electron.* (2011) 1081–1095.
- [6] Yanchi Zhang, et al., Small-signal modeling and modal analysis of wind turbine based on three-mass shaft model, *Electr. Power Components Syst.* 42 (7) (2014) 693–702.
- [7] Kamel Abboudi, et al., Dynamic behavior of a two-stage gear train used in a fixed-speed wind turbine, *Mech. Mach. Theory* 46 (12) (2011) 1888–1900.
- [8] Babak Badrzhadeh, S.K. Salman, Investigation of the torsional vibration and transient stability margin for doubly-fed induction generators, in: *Sustainable Alternative Energy (SAE), 2009 IEEE PES/IAS Conference on*, IEEE, 2009.
- [9] Jussi Sopanen, et al., Dynamic torque analysis of a wind turbine drive train including a direct-driven permanent-magnet generator, *Industrial Electron. IEEE Trans.* 58 (9) (2011) 3859–3867.
- [10] Andres E. Leon, Jorge Solsona, Sub-synchronous interaction damping control for DFIG wind turbines, *Power Syst. IEEE Trans.* 30 (1) (2015) 419–428.
- [11] E.V. Larsen, D.A. Swann, Applying power system stabilizers part ii: performance objectives and tuning concepts, *Power Apparatus Syst. IEEE Trans.* 6 (1981) 3025–3033.
- [12] L.C. Gross, Sub-synchronous grid conditions: new event, new problem, and new solutions, in: *Proc. Western Protective Relay Conf.*, 2010, pp. 1–5.
- [13] T. Ackermann, R. Kuwahata, Lessons learned from international wind integration studies, in: *AEMO Wind Integration WP4(A)*, Commissioned by Australian Energy Market Operator, 2011.
- [14] IEEE Committee Report, Reader's guide to subsynchronous resonance, *IEEE Trans. Power Syst.* 7 (1) (Feb. 1992) 150–157.
- [15] G.D. Irwin, A.K. Jindal, A.L. Isaacs, Sub-synchronous control interactions between type 3 wind turbines and series compensated AC transmission systems, in: *2011 IEEE Power and Energy Society General Meeting*, IEEE, 2011, pp. 1–6.
- [16] J. Adams, C. Carter, S.H. Huang, ERCOT experience with sub-synchronous control interaction and proposed remediation, in: *Transmission and Distribution Conference and Exposition (T&D), 2012 IEEE PES, IEEE, 2012*, pp. 1–5.
- [17] J. Ballance, S. Goldberg, Subsynchronous resonance in series compensated transmission lines, *IEEE Trans. Power Apparatus Syst.* 5 (PAS-92) (1973) 1649–1658.
- [18] L.A. Kilgore, D.G. Ramey, M.C. Hall, Simplified transmission and generation system analysis procedures for subsynchronous resonance problems, *IEEE Trans. Power Apparatus Syst.* 96 (6) (1977) 1840–1846.
- [19] R.J. Piwko, E.V. Larsen, HVDC system control for damping of subsynchronous oscillations, *IEEE Trans. Power Apparatus Syst.* 7 (1982) 2203–2211.
- [20] D.J. Kim, H.K. Nam, Y.H. Moon, A practical approach to HVDC system control for damping subsynchronous oscillation using the novel eigenvalue analysis program, *IEEE Trans. Power Syst.* 22 (4) (2007) 1926–1934.
- [21] K. Prasertwong, N. Mithulananthan, D. Thakur, Understanding low-frequency oscillation in power systems, *Int. J. Electr. Eng. Educ.* 47 (3) (2010) 248–262.
- [22] F. Zhu, H.G. Zhao, Z.H. Liu, et al., The influence of large power grid interconnected on power system dynamic stability, *Proc. CSEE* 1 (2007) 1–7.
- [23] T. Knuppel, et al., Small-signal stability of wind power system with full-load converter interfaced wind turbines, *Renew. Power Gener. IET* 6 (2) (2012) 79–91.
- [24] Ahmadreza Tabesh, Reza Iravani, Small-signal dynamic model and analysis of a fixed-speed wind farm-a frequency response approach, *Power Deliv. IEEE Trans.* 21 (2) (2006) 778–787.
- [25] Mohammad Hasanuzzaman Shawon, Ahmed Al Durra, S.M. Mueyen, Small signal stability analysis of fixed speed wind generator including SDBR, in: *Electrical Machines (ICEM), 2012 XXth International Conference on*, IEEE, 2012.
- [26] A. Tabesh, R. Iravani, Small-signal model and dynamic analysis of variable speed induction machine wind farms, *Renew. Power Gener. IET* 2 (4) (2008) 215–227.
- [27] Yateendra Mishra, et al., Small-signal stability analysis of a DFIG-based wind power system under different modes of operation, *Energy Convers. IEEE Trans.* 24 (4) (2009) 972–982.
- [28] J. Shanmuga Kani, et al., Small signal stability analysis of PMSG based WECS with coordinated controller tuning, in: *Green Computing Communication and Electrical Engineering (ICGCCEE), 2014 International Conference on*, IEEE, 2014.
- [29] Ren Wang, et al., Small signal stability analysis with high penetration of grid-connected wind farm of PMSG type considering the wake effect, in: *PES General Meeting Conference & Exposition, 2014 IEEE, IEEE, 2014*.
- [30] F. Wu, X.P. Zhang, P. Ju, Small signal stability analysis and control of the wind turbine with the direct-drive permanent magnet generator integrated to the grid, *Electr. Power Syst. Res.* 79 (12) (2009) 1661–1667.
- [31] S.M. Mueyen, R. Takahashi, T. Murata, et al., Transient stability analysis of permanent magnet variable speed synchronous wind generator, in: *Electrical Machines and Systems, 2007. ICEMS. International Conference on*, IEEE, 2007, pp. 288–293.
- [32] A.D. Hansen, G. Michalke, Modelling and control of variable-speed multi-pole permanent magnet synchronous generator wind turbine, *Wind Energy* 11 (5) (2008) 537–554.
- [33] P. Kundur, *Power System Stability and Control*, McGraw-hill, New York, 1994.
- [34] Y. Zhao, C. Wei, Z. Zhang, et al., A review on position/speed sensorless control for permanent-magnet synchronous machine-based wind energy conversion systems, *Emerg. Sel. Top. Power Electron. IEEE J.* 1 (4) (2013) 203–216.
- [35] J.S. Thongam, M. Tarbouchi, R. Beguenane, et al., An optimum speed MPPT controller for variable speed PMSG wind energy conversion systems, in:

- IECON 2012–38th Annual Conference on IEEE Industrial Electronics Society, IEEE, 2012, pp. 4293–4297.
- [36] K. Patil, B. Mehta, Modeling and control of variable speed wind turbine with permanent magnet synchronous generator, in: *Advances in Green Energy (ICAGE)*, 2014 International Conference on, IEEE, 2014, pp. 258–264.
 - [37] Shuhui Li, et al., Optimal and direct-current vector control of direct-driven PMSG wind turbines, *Power Electron. IEEE Trans.* 27 (5) (2012) 2325–2337.
 - [38] Jamal A. Baroudi, Venkata Dinavahi, Andrew M. Knight, A review of power converter topologies for wind generators, *Renew. Energy* 32 (14) (2007) 2369–2385.
 - [39] Zhenbin Zhang, et al., Encoderless model predictive control of back-to-back converter direct-drive permanent-magnet synchronous generator wind turbine systems, in: *Power Electronics and Applications (EPE)*, 2013 15th European Conference on, IEEE, 2013.
 - [40] Surya Santoso, Ha Thu Le, Fundamental time-domain wind turbine models for wind power studies, *Renew. Energy* 32 (14) (2007) 2436–2452.
 - [41] V. Akhmatov, H. Knudsen, A.H. Nielsen, Advanced simulation of windmills in the electric power supply, *Int. J. Electr. Power & Energy Syst.* 22 (6) (2000) 421–434.
 - [42] S.K. Salman, A.L.J. Teo, Windmill modeling consideration and factors influencing the stability of a grid-connected wind power-based embedded generator, *Power Syst. IEEE Trans.* 18 (2) (2003) 793–802.
 - [43] S.M. Muyeen, M.H. Ali, R. Takahashi, et al., Comparative study on transient stability analysis of wind turbine generator system using different drive train models, *Renew. Power Gener. IET* 1 (2) (2007) 131–141.
 - [44] G. Ramtharan, N. Jenkins, O. Anaya-Lara, et al., Influence of rotor structural dynamics representations on the electrical transient performance of FSIG and DFIG wind turbines, *Wind Energy* 10 (4) (2007) 293–301.
 - [45] N. Herisanu, V. Marinca, Gh Madescu, An analytical approach to non-linear dynamical model of a permanent magnet synchronous generator, *Wind Energy* 18 (9) (2015) 1657–1670.
 - [46] Joanne Hui, Alireza Bakhshai, Praveen K. Jain, Control and modeling of a wind energy system with a three-phase DCM boost converter and a sensorless maximum point power tracking method, in: *Transmission and Distribution Conference and Exposition (T&D)*, 2012 IEEE PES, IEEE, 2012.